Andreas Manalo Bergli - 8
Davyd Lebedovskyi - 501
Richard Hansen - 249
Somjate Srirabai - 19

Emnekode: BAO302

Emnenavn: Bachelorprosjekt

Oppdragsgiver: Statkraft

Innleveringsdato: May 15, 2025

Antall sider: 55

Antall ord: 11,971

Tilgjengelighet: Fri ☑ Begrenset ☐

School of Economics, Innovation, and Technology

Kristiania University of Applied Sciences, Oslo, Norway

Benchmarking Maintenance Efficiency Using Machine Learning:
A Case Study at Statkraft's Hydropower Plants

Benchmarking av vedlikeholdseffektivitet ved bruk av maskinlæring:
En casestudie av Statkrafts vannkraftverk

Kristiania

Våren 2025

Abstract

This report presents a solution based on Machine Learning (ML) for benchmarking the maintenance efficiency of **Scandinavian hydropower plants**, with a focus on **Statkraft AS**, a leading renewable energy company. The goals were to assess plant efficiency by analyzing historical maintenance data, particularly planned and forced downtime. Then leveraged **machine learning models to predict maintenance-based efficiency (MBE) scores**, linking historical data to operational performance.

The **report begins with an overview of hydropower operations**, turbine types, and maintenance strategies, laying the foundation for understanding how maintenance impacts efficiency. Benchmarking was explored as a key tool for comparing plant performance and identifying best practices. The concept of **MBE** was proposed to connect maintenance efforts directly to operational outcomes.

Due to the absence of established efficiency scores, **synthetic data was created** using an average of 2-year real data as seed. After thorough preprocessing, we tested four non-linear regression architectures: **XGBoost, Random Forest, AutoEncoder + regressor, and a Multi-Layer Perceptron (MLP)**. The experiments were conducted using a 50-trial random hyper-parameter search and 10-fold cross-validation. The metrics: RMSE, MAE, and R^2, were used to evaluate the models' performance.

Experimentation results: All models achieved high explanatory power on the synthetic hold-out set with $R^2 \geq 0.9938$. However, Random Forest Regressor performed best, having achieved: **RMSE 0.96, MAE 0.19 and R^2 0.9989**, despite moderate train–test divergence compared to other models. The trained RandomForest was applied to the imputed "Real" dataset and generated "Predicted Efficiency Score" ranging between 0-1.

The predicted efficiency scores was then integrated into a Power BI dashboard, aggregated results by region, asset-class and cost-generating unit, to highlight under-performers as well as identify high-performers. This **allows Statkraft to compare maintenance effectiveness across its plants**, quantify opportunity losses, and prioritize remedial actions.

Key deliverables includes the **Power BI dashboards** and **machine learning code** were handed over with full documentation to support future development.

Future improvement: While acknowledging that the reliance on synthetic data was a limitation, the project succeeded in meeting its core objectives and demonstrated clear potential for real-world application. Future improvements include **refining synthetic data** realism and possible use of SCADA-data for an integrated early preventative maintenance detection and warning.

Acknowledgments

This bachelor thesis would not have been possible without the generous support, collaboration, and guidance of numerous individuals, to whom we are sincerely grateful.

We would like to express our heartfelt thanks to **Inger Marie Solbrekke**, Vice President of Digital Finance Nordics, and **Mauricio Ochoa Vasquez**, Senior Digital Financial Advisor for their invaluable assistance in providing access to key data sources and strategic context throughout this project.

Special appreciation is extended to our supervisors: **Guru Bhandari** (Ph.D., Kristiania University College) for his insightful academic supervision and continuous encouragement, and **Mauricio Ochoa Vasquez** at Statkraft, for his external guidance and practical perspective, which significantly enriched the relevance and quality of our work.

We are also grateful to members of the analytics and asset performance teams. **Nicolas Parow**, Senior Asset Performance Analyst, and **Henrik Vassdal**, Asset Performance Analyst, offered detailed insights and data support that were critical in shaping the analysis—particularly with SCADA-data and understanding of downtime instances and performance metrics.

Our thanks also go to **Rune Hansen**, Vice President Co-owners and Transmission, for helping us understand the nuance difference between the terms used, which greatly enhanced our understanding of the operational environment.

We would like to acknowledge the contributions of the IT department for their assistance in setting up and maintaining the computing environment used throughout the project. Their technical support ensured uninterrupted access to the tools and infrastructure required for data analysis and model development.

Finally, we are thankful to all individuals—named and unnamed—who supported us through discussions, clarifications, and resources during this process. Your contributions were instrumental in helping us complete this work, and we are deeply appreciative.

Data Science Student Group 94

Table of Contents

1 Introduction

Hydropower plant operation is inherently complex, requiring deep knowledge of both power generation technologies and maintenance strategies to ensure efficiency and reliability. As renewable energy becomes increasingly essential to global energy systems, optimizing hydropower performance has become a critical goal for sustainable development.

This report explores both theoretical and practical frameworks for improving hydropower plant performance, with a focus on maintenance strategies, performance benchmarking, and Machine Learning (ML) applications. It begins with an overview of hydropower generation fundamentals, including turbine types and their maintenance needs. Historical maintenance data plays a central role in this study, forming the foundation for predictive and prescriptive maintenance through effective data extraction, preparation, and exploration.

A key concept is Maintenance-Based Efficiency (MBE), which links maintenance practices directly to performance outcomes. Benchmarking is also examined as a tool for comparing operational efficiency across facilities.

The report then presents ML techniques—XGBoost Regressor, Random Forest Regressor, MLP, and AutoEncoder combined with a Regressor—to identify patterns in maintenance data and anticipate inefficiencies. These models enable data-driven decision-making and proactive maintenance planning. Post-training validation techniques are addressed to ensure model reliability in real-world scenarios.

By integrating hydropower maintenance knowledge with advanced analytics, this report aims to develop a diagnostic tool for benchmarking and identifying best practices.

To ground the analysis, the report begins with an overview of the project at its core, focusing on the application of MBE in hydropower operations. It also introduces Statkraft AS, a leading renewable energy company and key collaborator on the project, followed by a review of the theoretical foundations guiding the study.

1.1 The project: MBE for Hydropower Plants

Problem Statement: A case study leverages ML, grounded in an analysis of historical maintenance records from Statkraft's hydropower plants, to develop a MBE-score aimed at enhancing benchmarking capabilities.

Scope: This project focuses on Statkraft's hydropower plants located in Norway and Sweden, using data from 2021-2024.

Primary Goals: The primary goals of this project are:

- To provide maintenance based analysis on the historical data from 2021-2024.
- To develop a ML model that quantifies the efficiency of hydropower plants based on their maintenance performance.
- To provide a tool to compare maintenance based efficiency across the hydropower plants.
- To benchmark maintenance efficiency across plants, business groups, and regions.

1.2 The company: Statkraft AS

Statkraft, headquartered in Oslo, Norway, is fully owned by the Norwegian government through the Ministry of Trade, Industry, and Fisheries. As Norway's largest energy producer and Europe's leading generator of renewable energy, Statkraft plays a key role in the transition to a sustainable energy future. Founded in 1992, the company operates in over 20 countries across Europe, Asia, and South America. According to its 2024 Annual Results, Statkraft's total installed capacity reached 22.3 gigawatts (GW) [27].

Global Presence: Statkraft has a strong presence in Europe, especially in Norway, Sweden, Germany, and the UK, where it is a major player in renewable energy generation. The company has heavily invested in wind and solar power, particularly in Spain and the Nordic region [28].

In South America, Statkraft has expanded its renewable operations in Brazil, Chile, and Peru, focusing on wind and solar energy over hydropower [29]. In Brazil, it has become a key player in wind energy [30]. In Asia, the company has focused on hydropower and solar projects in countries like India and Nepal, though recent divestments in India reflect a strategic shift toward markets with higher growth potential [31].

Hydropower Operations: Statkraft maintains one of Europe's largest and most sustainable energy portfolios, with hydropower at its core—accounting for over 60% of the company's total energy generation. It operates more than 150 hydropower plants, primarily in Norway and Sweden.

In Norway, Statkraft owns 123 hydropower plants with a total installed capacity of 11,452 MW, significantly contributing to Europe's reservoir capacity. In Sweden, the company operates 54 plants, generating approximately 5.4 TWh annually. Notable facilities include Svartisen (350 MW) in Norway, and Skeen (5 MW) and Nissaström (11 MW) in Sweden [32].

1.3 Theoretical Background

Theoretical frameworks essential for optimizing hydropower plant performance—including power generation technologies and methodologies for ensuring maintenance and operational efficiency—are reviewed, with a focus on maintenance strategies, ML applications, and performance benchmarking.

The concept of MBE is also introduced, highlighting its role in enhancing maintenance practices and improving plant performance. Benchmarking is also explored as a critical tool for evaluating efficiency across various operational contexts.

Additionally, the application of ML models—including XGBoost Regressor, Random Forest Regressor, MLP, and AutoEncoder with a Regressor, to analyze maintenance data, is explored.

This background knowledge lays the foundation for understanding how advanced data-driven strategies can optimize the performance and reliability of hydropower plants.

1.3.1 Hydropower Plants Basics

Hydropower plants generate electricity by converting the kinetic energy of flowing or falling water into mechanical energy via a turbine, which is then converted into electrical energy by a generator. The process depends on three key components: water flow, a turbine, and a generator [8], [33].

Hydropower Systems and Maintenance: While hydropower plants are designed for long-term efficiency, regular maintenance is crucial for optimal performance. Neglecting upkeep can result in costly repairs, unplanned downtime, and reduced energy production. Preventive and corrective maintenance strategies help identify issues early, extend the lifespan of key components like turbines and generators, and maintain maximum energy output.

1.3.2 Types of Turbines and Maintenance Needs

Different types of turbines are used in hydropower plants, primarily classified as impulse and reaction turbines, each adapted to specific operating conditions, as outlined below.

- **Impulse Turbines:** Designed for high-head applications—where water falls from significant heights, generating energy through gravity, and typically has low flow rates. Impulse turbines convert the water's kinetic energy into mechanical energy through direct impact [9].

  Pelton Turbines: Pelton is a type of Impulse Turbine that use spoon-shaped buckets mounted around movable wheel called "runner" to capture water jets. These runners and the "buckets" are the primary maintenance subject in Pelton turbines, requiring regular inspections for erosion and cavitation damages, as well as sharpening the bucket to reduce back-splash [24]. Figure 1 illustrates the basic structure of a Pelton turbine.

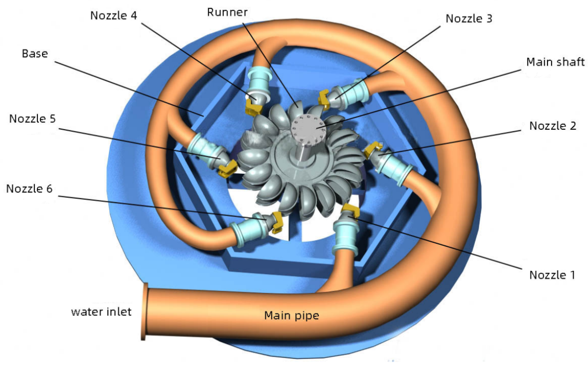


Figure 1: Pelton Turbine

Source: https://www.huahydro.com/2014/10/24/pleton-turbine/

- **Reaction Turbines:** This type of turbine is suitable for low- to medium-head with continuous water flow conditions. Reaction turbines generate power through both the pressure and kinetic energy of water.

  Francis turbine is a widely used reaction turbine, accounting for over 60% of global hydropower generation. It operates efficiently under various conditions, with key components including a spiral casing, stay vanes, and a runner with fixed blades. Maintenance focuses on inspecting the runner and guide vanes for wear and cavitation, as well as ensuring proper alignment and lubrication of moving parts [7]. Figure 2 shows Francis turbine's basic structure.

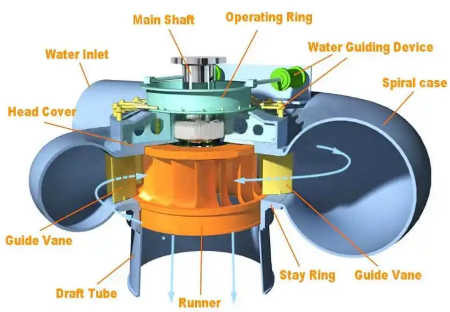


Figure 2: Francis Turbine

Source: https://theconstructor.org/practical-guide/francis-turbines-components-application/2900/

Kaplan turbine, developed by Viktor Kaplan in 1913, is a propeller-type reaction turbine with adjustable blades, ideal for low-head, high-flow conditions. Figure 3 outlines Kaplan turbine's basic structure. Its design ensures efficient operation across various flow scenarios, with efficiencies often exceeding 90% [12]. Routine maintenance involves inspecting runner blades and guide vanes for abrasion and cavitation, as well as checking critical areas like blade pivots, control rods, and piston rods for fatigue cracks [19], [23].

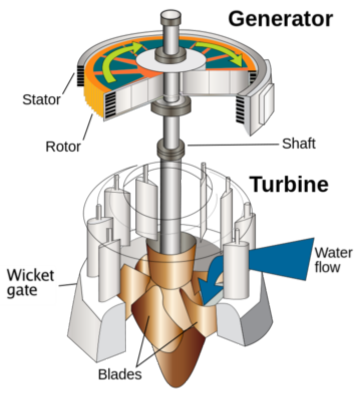


Figure 3: Kaplan Turbine

Source: https://www.huahydro.com/2014/10/24/pleton-turbine/

1.3.3 Maintenance Activity Types:

Maintenance activities are broadly categorized into two main types: Planned Maintenance and Forced Maintenance. **Planned Maintenance or Preventative** is a proactive approach that schedules maintenance tasks in advance to prevent disruptions and equipment failure, and optimize uptime. It includes:

- Condition-Based Monitoring (CBM): It involves monitoring equipment conditions through real-time data or periodic checks. Maintenance is performed only when specific indicators signal declining performance or potential failure, minimizing downtime by ensuring that maintenance or interventions are carried out only when necessary [34].

- Condition-Based Repair (CBR): This approach builds upon condition monitoring by initiating repairs when specific condition thresholds are met, aiming to fix only what is necessary based on the equipment's real-time performance data [14].

In contrast to preventive maintenance, **Forced Maintenance or Corrective** is reactive, triggered by unexpected equipment failures. It often causes operational disruptions and requires immediate repairs or part replacements, which may not be readily available, making it more costly and disruptive [5]. Though not ideal, corrective maintenance is sometimes unavoidable and can lead to prolonged, unpredictable downtime. Implementing effective maintenance strategies helps reduce its frequency and impact [37].

1.3.4 Benchmarking: A Key Concept in Performance Measurement

Benchmarking is the process of comparing organizational performance against industry standards or competitors to identify areas for improvement [6]. It involves setting performance metrics, evaluating current practices, and analyzing leading organizations to establish benchmarks for success [2]. This helps identify best practices that can improve efficiency and competitiveness.

Benchmarking types include internal (comparing departments within the same organization), competitive (against direct competitors), and functional (against leaders in similar functions across industries) [10]. Overall, benchmarking supports continuous improvement, operational efficiency, and innovation.

ML Generated Efficiency Score in Benchmarking: To enhance benchmarking, hydropower plants can use efficiency scores derived from method tools and ML. These objective metrics help assess performance, identify gaps, and guide resource optimization. Integrating ML-based scores into benchmarking enhances operational efficiency, sustainability, and competitiveness—making it particularly valuable for evaluating renewable energy projects across diverse regions. [1], [3].

Recent studies highlight the effectiveness of ML in predicting and optimizing hydropower plant performance. For instance, a study on the Hirfanlı Dam used hybrid models combining tree-based methods (Random Forest, Gradient Boosting) with deep learning (LSTM) to forecast power production. The models achieved high accuracy, with a correlation coefficient of 0.977 and R^2 of 0.955, demonstrating strong potential for reliable performance evaluation and optimization [41], [22].

By integrating ML-generated efficiency scores into benchmarking practices, hydropower plants can achieve more accurate and dynamic assessments of their performance, leading to informed decision-making and continuous improvement.

1.3.5 Hydropower Plant Efficiency Scoring.

Evaluating hydropower plant efficiency is challenging due to variations in environmental and technical setups. Differences in water sources (rivers, dams, waterfalls), plant size, and turbine types lead to distinct operational characteristics, making direct comparisons difficult.

Despite these complexities, hydropower plant efficiency can still be evaluated using proxy measures such as overall efficiency and capacity factor. More technical metrics, like turbine and generator efficiency, aid in comparison, while economic efficiency can be assessed by relating maintenance or operational costs to revenue or output [36].

Turbine Efficiency: Refers to how effectively the turbines convert water energy into mechanical power. This can be influenced by factors like the design and age of the turbine, the operational conditions (such as water flow and pressure), and the regularity of maintenance practices [36].

Generator Efficiency: It measures the conversion of mechanical power into electrical energy. If turbines and generators are operating below their optimal efficiency, energy losses can increase, directly impacting the plant's overall performance [36].

Maintenance-Based Efficiency: The cost of repairing or upgrading aging equipment, and the return on investment from different energy market conditions [18].

1.3.6 Maintenance-Based Efficiency

MBE assesses the relationship between maintenance activities and a hydropower plant's operational performance [39]. It evaluates how maintenance practices and costs impact energy output, lifespan, and reliability, highlighting their role in sustaining or improving overall efficiency [17]. MBE encompasses:

- Direct Maintenance Costs: These include the expenses related to routine and corrective maintenance, repairs, and equipment replacements.
- Indirect Maintenance Benefits: These are the advantages of minimizing downtime, preventing major failures, extending equipment lifespan, and ensuring optimal energy production.
- Opportunity Losses: This includes the cost of lost energy production due to maintenance-related outages or inefficiencies, and the broader opportunity cost of not utilizing plant capacity during periods of potential high energy demand or pricing.

In essence, MBE not only measures how well a plant maintains its key systems (e.g., turbines, generators) to ensure maximum productivity but also assesses the economic impact of lost opportunities due to maintenance shortcomings. A well-maintained system reduces both direct and indirect losses, ensuring reliable operation with minimal downtime and maximizing energy and financial returns.

Key Components of Maintenance-Based Efficiency:
The efficiency of maintenance in a hydropower plant is determined by the relationship between maintenance costs and energy production. High maintenance costs with minimal performance gains indicate inefficiencies, while low costs with high energy production suggest effective resource use. Long-term efficiency is also linked to reliability and minimal downtime, with regular preventive maintenance crucial in avoiding costly breakdowns and disruptions to power generation [17].

Adopting data-driven maintenance strategies enhances long-term efficiency by tracking and improving plant performance throughout its life cycle [16]. By minimizing downtime and reducing failure risks, plants can achieve short-term energy gains and long-term sustainability, optimizing the balance between maintenance costs and power generation.

Finally, effective asset management is vital for ensuring that the plant operates efficiently over time. This involves closely monitoring the life cycle of each piece of equipment, managing spare parts, and scheduling maintenance activities at the optimal time. Such proactive management helps avoid unnecessary costs, prevent equipment failures, and ultimately ensure the plant runs smoothly and efficiently. [20]

1.3.7 Machine Learning Models

XGBoost (Extreme Gradient Boosting) Regressor is a powerful ML model based on gradient boosting frameworks. It is designed to be highly efficient, scalable, and accurate, making it particularly well-suited for structured data problems such as maintenance forecasting in hydropower systems, due to the tabular nature of data in such tasks. The core idea behind XGBoost is to build an ensemble of weak prediction models, typically decision trees, in a sequential manner where each new tree corrects the residual errors of the previous ones [40].

Key Architecture and Assumptions

- **Regularized Learning Objective:** Unlike traditional gradient boosting, XGBoost includes L1 (Lasso) and L2 (Ridge) regularization to control model complexity and prevent overfitting.

- **Parallel Processing:** It supports parallelization of tree construction, which significantly improves training speed, especially on large datasets.

- **Tree Pruning:** Uses a more sophisticated approach known as "depth-first" pruning, it builds each tree to a maximum depth then walks back removing branches that do not contribute significantly to performance.

- **Weighted Quantile Sketch:** For finding the best split, it efficiently handles weighted data, which is useful when certain records are more important.

These key architectural features are visualized within Figure 4.

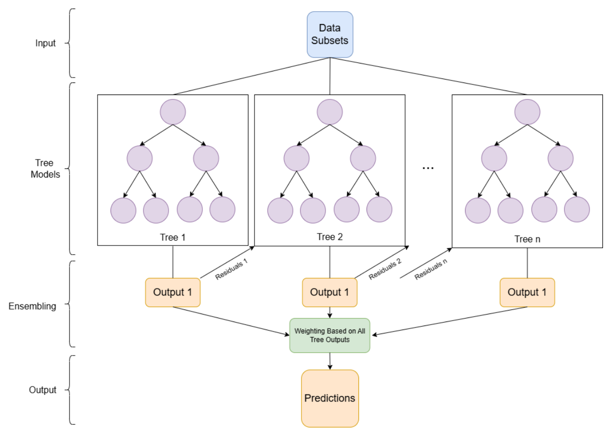


Figure 4: Depiction of a standard XGBoost architecture.

Model Assumptions and Handling of Data

- Distribution Assumptions: XGBoost makes no strong assumptions about the distribution of the input features or the target variable. This flexibility allows it to perform well on a wide range of regression tasks.

- Handling of Missing Values: One of XGBoost's strengths is its built-in ability to handle missing values. During training, the algorithm learns the best direction (left or right in the tree) to take when a value is missing, which can be particularly useful in maintenance datasets where sensor or log data might be incomplete.

- Feature Scaling: Feature scaling (e.g., normalization) is generally not required, as XG-Boost is based on decision trees that are invariant to monotonic transformations of input features.

XGBoost's balance of accuracy, speed, and robustness to missing or noisy data makes it a compelling choice for predictive maintenance tasks in hydropower systems, especially when interpretability and real-time performance are important.

The Random Forest Regressor is an ensemble learning method that constructs multiple decision trees during training and outputs the mean prediction of the individual trees to improve predictive accuracy and control overfitting. It is a versatile and widely-used model in regression tasks, including predictive maintenance in hydropower systems [25], [21].

Key Architecture and Assumptions

- **Ensemble Learning:** Builds an ensemble of decision trees, each trained on a random data subset, and averages their predictions to improve accuracy and reduce overfitting.
- **Bootstrap Aggregating (Bagging):** Each tree is trained on a bootstrap sample (random sampling with replacement) of the data, allowing for diversity among the trees.
- **Feature Randomness:** At each split in a tree, a random subset of features is considered, promoting diversity and decorrelating the trees.
- **No Strong Assumptions:** The model does not assume any specific distribution of the input features or the target variable, making it flexible and robust to various data types.
- **Scalability:** Each tree is fully independent, so training scales nearly linearly with the amount of cores used. Many libraries (e.g scikit-learn) exploit this to enhance performance [26].

These assumptions are visualized in Figure 5.

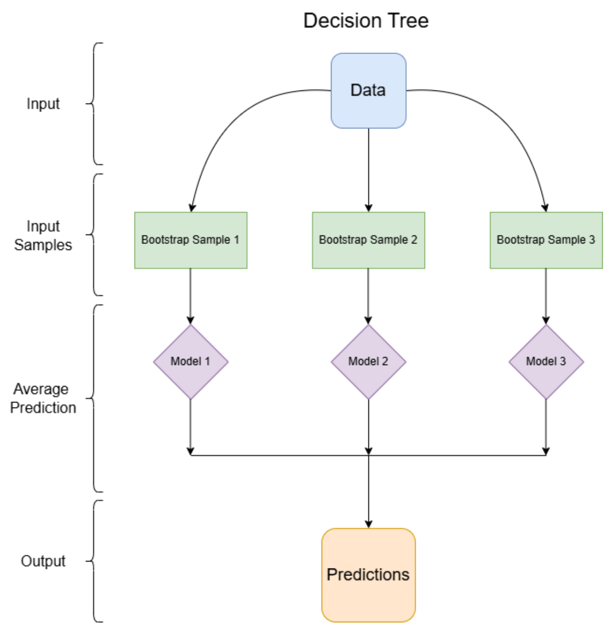


Figure 5: Depiction of a standard Random Forest Regressor architecture.

Handling of Data

- **Missing Values:** Random Forest Regressor does not inherently handle missing values. It is recommended to preprocess the data to handle missing values appropriately before fitting the model.

- **Feature Scaling:** Feature scaling is not required, as decision trees are invariant to monotonic transformations of the input features.

- **Data Representation:** The input data should be provided as a NumPy array or a Pandas DataFrame. Categorical variables should be encoded appropriately, such as using one-hot encoding.

The Random Forest Regressor's combination of robustness, speed, flexibility, and interpretability makes it a powerful tool for predictive maintenance tasks, such as forecasting failures in hydropower plants.

AutoEncoder with Regressor is a hybrid model that combines an unsupervised autoencoder with a supervised regression model, enabling it to learn compressed representations before performing predictions [35], [38].

Key Architecture and Assumptions

- **Bottleneck Compression:** forces information through a narrow latent layer via encoding then reconstructs them via decoding, distilling only the most salient nonlinear features [15].

- **Multi-Layered Architecture:** an unsupervised autoencoder is first trained to denoise inputs and extract salient features; its compressed latent embeddings are then passed to a regressor that learns to map those features to the target.

- **Switching out Decoder for Regressor:** Instead of reconstructing the input data, the decoder is replaced with a supervised regression head that maps the latent representations directly to the target variable, aligning the learned features with the prediction task.

A diagram depicting an AutoEncoder with a regresson layer is visulized in Figure 6 where Outputs from the Latent Space is directly fed into the first Regressor hidden layer.

Handling of Data

- **Latent Sufficiency:** assumes the compressed features retain all the most relevant information.

- **Input Normalization:** Assumes normalized, and continuous data for stable gradient-based training. Without scaling, features with larger magnitudes can dominate the loss and skew the learned embeddings.

- **Missing Values:** Missing values must be imputed or masked before feeding into the autoencoder.

- **No Strong Assumptions:** no strong assumptions on input distributions are made, the predictions come from discovering whatever structure is present. This flexibility lets the model adapt to skewed, multi-modal, or non-Gaussian data.

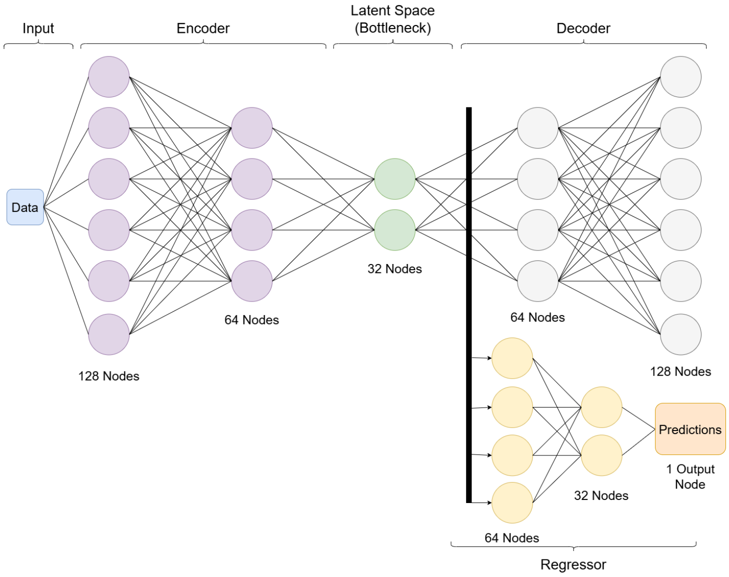


Figure 6: Architecture of an AutoEncoder with a regresson Layer.

A MLP Regressor is a type of feedforward neural network that learns a nonlinear mapping from inputs to a continuous target via backpropagation. It is particularly useful for modeling complex, non-linear relationships in predictive maintenance data.

Key Architecture and Assumptions

- **Layer Stacking & Activations:** Successive fully connected layers apply a linear transform (W x + b) followed by a nonlinear activation (e.g. ReLU or tanh). This lets each layer take the patterns discovered by the previous layer and build on them to learn progressively more complex features. [11]

- **Backpropagation:** Uses gradient descent to updates weights during training. optimizers such as Adam or RMSprop adapt learning rates per parameter and incorporate momentum for smoother convergence.

- **Regularization:** Adds Weight Decay (L2/Ridge) to the loss. This shrinks all weights towards zero, discouraging features from dominating resulting in more stable outputs. Further penalties (L1/Lasso) encourages exact zeros in the weight vector, effectively performing feature selection by driving less-important inputs out of the model.

A diagram depicting a standard MLP model is visulized in Figure 7.

Handling of Data

- **Categorical Encoding:** Requires numerical inputs; categorical features must be encoded.

- **Input Scaling:** All inputs needs to be normalized or standardized so the gradients have a balanced magnitude.

- **Missing Values:** Missing values must be imputed or masked before feeding into MLP models.

- **No Strong Assumptions:** no strong assumptions on input distributions are made, the

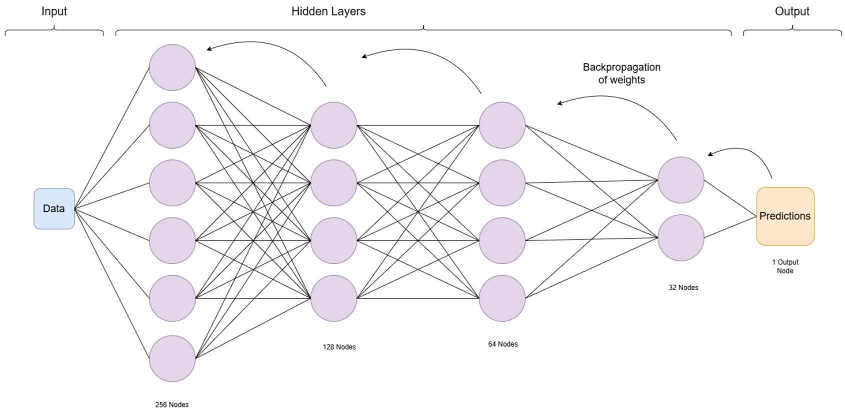


Figure 7: Depiction of a standard MLP architecture.

predictions come from discovering whatever structure is present. This flexibility lets the model adapt to skewed, multi-modal, or non-Gaussian data.

2 Methodology

This section outlines the approach for developing and evaluating ML models to predict hydropower plant efficiency. It covers considerations like data sensitivity, IT constraints, project scope, synthetic data creation, preprocessing, model selection, performance evaluation, and communication practices for effective collaboration.

2.1 Data Sensitivity and IT Issues

Data security is crucial at Statkraft, given its role in managing Norway's critical infrastructure. Stringent measures ensure protection, with secure personal computers and only pre-approved software allowed. Data cannot be transferred outside secure networks, and all processing occurs within controlled frameworks. In line with these security protocols, identifiable information about power plants has been anonymised.

2.2 Data Requirements

To support the development of efficiency prediction models, the following data will be required:

- A list of power plants along with associated metadata (e.g., location, capacity, type).
- Historical maintenance records detailing what maintenance was performed and when.
- Production volume data from each power plant over a time frame of interest.
- The record of the plants' state "available" or "unavailable" for operation.
- Historical maintenance expenditure data for each plant.

2.3 Use of Synthetic Data

A key challenge is the lack of a defined "ground truth" for hydropower plant efficiency, complicating the validation of predicted scores. To address this, we use synthetic data with known efficiency relationships to train and evaluate regression models, ensuring they learn meaningful patterns. Synthetic data allows consistent testing of model performance, stability, and generalization, supporting reliable validation and iterative refinement [4].

2.4 Data Preprocessing

Data preprocessing is a fundamental stage in any data-driven project, ensuring that raw data is transformed into a clean, consistent, and structured format suitable for model development. The preprocessing pipeline consists of a series of interdependent steps, beginning with error correction and ending with model-specific transformations. The key components of the data preprocessing workflow includes: data cleaning, handling missing or inconsistent values, feature engineering, and model-level preprocessing.

Data Cleaning: Data cleaning is the process of handling issues such as duplicates, and outliers, which may distort the results. This step ensures that the dataset is consistent, reliable, and ready for further analysis [13].

Handling missing/inconsistent values: Data quality was assessed by detecting naming discrepancies, missing values, and inconsistencies in data types or units. The extent of missing data was quantified and visualized to guide the selection of appropriate imputation strategies—ranging from simple statistical methods to advanced ML approaches, or record exclusion, depending on the nature of the missingness.

Feature Engineering: In this step, the need for creating new features from existing ones or transforming current features into more meaningful representations is assessed. These modifications are intended to optimize model learning and improve predictive performance.

Model level Preprocessing Right before using the data in a model, it is important to make sure we have the correct transformation, or use scaler if needed. Encode categorical features if they are going to be used by the model. Deciding on the train-test split, if not dealt with otherwise. Now the data should mostly be ready to be used in the ML model.

2.5 Model Selection

Model selection begins with identifying the problem type and the nature of the data. As the objective is to predict a continuous efficiency score from operational and performance features, the task falls under supervised learning. Given the continuous target variable, regression models are selected over classification approaches.

While unsupervised learning techniques, such as clustering, can provide useful insights into data structure, groupings, or anomaly detection, they fall short when the objective is to produce quantifiable, interpretable efficiency scores. Clustering can suggest relative groupings but does not yield explicit measures that can be effectively used for benchmarking or performance ranking across plants.

In contrast, supervised regression models enable the generation of continuous efficiency scores that are not only predictive but also allow for meaningful comparison and targeted decision-making. As a result, our model selection process will focus on evaluating a range of supervised regression algorithms; including tree-based methods, linear models, and deep learning approaches, to identify the most accurate and reliable models for performance assessment and benchmarking.

2.6 Evaluation Metrics and Validation Methods

To evaluate the performance of the predictive models and ensure robustness, we employed three commonly used regression metrics: **Root Mean Square Error (RMSE)**, **Mean Absolute Error (MAE)**, and the **Coefficient of Determination ($\mathbf{R^2}$)**.

Root Mean Square Error (RMSE) measures the square root of the average squared differences between the predicted and actual values. It penalizes larger errors more heavily, making it suitable when large deviations are particularly undesirable:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Mean Absolute Error (MAE) is the average of the absolute differences between predicted and actual values. Unlike RMSE, it treats all errors equally, offering a more balanced perspective when outliers are not a major concern:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

R² Score (Coefficient of Determination) reflects the proportion of variance in the target variable that can be explained by the model. It ranges from 0 to 1, where 1 indicates perfect prediction:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

MAE calculates the average magnitude of the errors in a set of predictions, providing a straightforward measure of average model error. RMSE, on the other hand, squares the errors before averaging and then takes the square root, which gives higher weight to larger errors.

In contrast, the R^2 score (Coefficient of Determination) measures the proportion of variance in the target variable that the model can explain. An R^2 value closer to 1 indicates a better fit, with 1 representing a perfect prediction.

2.7 Communication Methods

To ensure clear and consistent communication, the group appointed Andreas Manalo Bergli as the designated contact person. All correspondence with external stakeholders, including the internal advisor, is coordinated through him. This approach helps avoid confusion, maintains a unified message, and streamlines the flow of information between the group and relevant parties.

Communication with Kristiania: We established a routine of bi-monthly progress updates with our internal advisor via email to kept the school informed of the progression of the project, and to maintain open and transparent communication throughout the development process.

Communication with Statkraft: Regular sprint meetings were scheduled with our external supervisor from Statkraft. Initially, the meeting was held once a week. As more deliverables were being worked on simultaneously, the frequency was increased to twice a week. This is to ensure a close monitoring of progress and timely feedback, allowing more frequent check-ins to keep the projects on track. As time and projects progressed and tasks began to slow down, the meeting frequency reverted back to once a week.

Communication within the Bachelor group: The team uses Microsoft Teams for internal communication and task coordination. A dedicated group chat enables real-time discussions, updates, and issue resolution, while a separate chat with the external supervisor ensures transparent communication of key information and progress. Shared documents are also used to maintain to-do lists and track ongoing responsibilities, helping the team stay aligned and organized throughout the project.

3 Analysis and Development:

The section outlines the process of transforming data into actionable insights. It covers requirements clarification, data collection, and exploratory data analysis (EDA) to understand the dataset. The section also details the application of ML models for efficiency prediction and the tools used throughout the process.

3.1 Requirements Clarification

To ensure a clear and shared understanding of the project scope, we conducted a series of meetings with key stakeholders, including the Senior Business Performance Advisor and the Vice President of Digital Finance Nordics. These discussions were essential in clarifying expectations and aligning the project's objectives with the strategic interests of the organization.

Given the stakeholders' domain expertise in the Nordic energy market, the project scope was narrowed to focus specifically on **hydropower plants** located in **Norway and Sweden**.

The final scope was shaped by a combination of regional focus and the availability of relevant historical data. These early decisions helped ensure that the work remained both feasible and aligned with business objectives.

3.2 Data Collection.

The foundation of any ML project lies in the quality and relevance of the data used. In this project, the initial phase focused on collecting, validating, and understanding data from a range of internal systems. Our goal was to build a model that evaluates hydropower plant efficiency using maintenance and operational records. However, the path from concept to clean, usable data proved complex and iterative.

Initial Considerations and Shift in Approach
At the project's outset, stakeholders highlighted Statkraft's need for power plant comparison and benchmarking. Since efficiency scores were unavailable, we first explored unsupervised learning, but validation without labeled data proved challenging. We then shifted to supervised learning, generating synthetic data with known efficiency scores for controlled training before applying the model to real-world data.

We also considered modeling specific components, such as generators and turbines, but ultimately discarded this due to inconsistent, detailed data and the added complexity of tracking component-level metrics across plants.

Initially, we thought about incorporating features from Supervisory Control and Data Acquisition (SCADA) systems, but this approach was hindered by challenges with data accessibility, complexity, and inconsistencies across plants. Given the significant variations in water sources, turbine configurations, and installed capacity, creating a uniform dataset for meaningful comparison proved difficult. Ultimately, due to time constraints and limited data, SCADA data was excluded from the models.

Data Sources Utilized

To create a robust training dataset, data were gathered from multiple queries including:

- **SAC Maintenance:** Provide details on maintenance incidences and accompanying details as well as associated costs.
- **Mapping Table:** Contained plant identifiers (multiple system of plant identification system) and grouping hierarchies.
- **SAC Cost Øre/KWH** Offered production based financial information.
- **SAC Market Adjusted Availability** Included downtime types (Planned VS Forced), its duration and their market adjusted equivalents. Market Adjusted figure is
- **SAC Brady Production** Supplied the actual and target production figures.

These datasets were accessed via internal business systems such as SAP and SAC, often requiring coordination with domain experts to interpret unfamiliar terms and validate the meaning of various features.

Data Collection Challenges Throughout the data collection process, several challenges emerged. Some of the most notable include:

- **Identifier Inconsistencies:** One of the most critical challenges was the absence of a consistent key for linking datasets. Across different queries, there appeared to be at least four distinct methods for identifying power plants; each likely tailored to the needs of specific departments or systems. However, none served as a universal or authoritative plant identifier. Significant effort was required to establish a reliable identifier that could consistently bridge data across these disparate sources.
- **Data Quality Issues:** We encountered missing values, duplicate records, and inconsistent time formatting. Correcting data entry errors and reconciling varying units of measurement were essential steps in the cleaning process. Some that needed manual correcting.
- **Terminology Differences:** Many datasets used different terms for the same concept, especially when spanning regions or departments. For example, plant names varied due to translation differences or local naming conventions.
- **Access Limitations:** We were not able to directly query or extract the data ourselves in many cases, relying instead on scheduled exports or support from system experts.
- **Understanding the Data:** Several working sessions and meetings were required to clarify the purpose and structure of various potential features. This included deciphering domain-specific jargon and understanding business logic embedded in calculated fields.

3.3 Exploratory Data Analysis (EDA)

Before modeling or drawing conclusions, a thorough exploration of data is essential. EDA uses statistical techniques and visualizations to understand data structure, variable relationships, and trends. In this project, EDA was performed on historical maintenance data from hydropower plants (2021–2024), focusing on both planned and unplanned maintenance.

The analysis aimed to uncover patterns and trends across Power Plant Groups (PPGs), Regions, Cost Generating Units (CGUs), and individual plants. Using statistical techniques and visualizations, EDA provided insights into the dataset's structure, guiding the selection of analytical methods and supporting the goal of optimizing maintenance costs.

In addition to the standard time scales such as years and months, several key grouping concepts are essential for understanding the structure of the data and the subsequent analysis. These groupings provide important context for interpreting patterns and trends across the hydropower plants. The following section outlines these critical concepts.

- **Regions:** Hydropower plants in Nordic regions are categorized into four geographical regions: Northern Norway, Central Norway, Southern Norway, and Sweden (Statkraft does not have operations in other Nordic countries.)

- **Power Plant Groups (PPG):** Within each "Regions", there can be multiple PPGs.

- **Cost Generating Unit (CGU):** Within each PPG, there can be multiple CGUs, which refer to groups of hydropower plants that are closely interconnected through shared water intake. Specifically, a CGU consists of a series of consecutive plants along a shared waterway, where the outflow from an upstream plant serves as the inflow for the subsequent plant.

- **Asset Class:** All power plants are categorized into three classes based on their production contribution and criticality to the overall operation of Statkraft. These classes are color-coded as follows: 1 = blue, 2 = yellow, and 3 = red, with red representing the most critical level (this colors scheme is used throughout visualization within Statkraft. e.g. See Figure 22.)

- **Maintenance Activity Types:** There are four categories of maintenance activities, each with associated costs: Preventive, Corrective, Modification, and Operation. Preventive maintenance is further categorized into Condition-Based Repair, Condition Monitoring, and Periodic Overhaul, while Corrective maintenance is divided into Planned Corrective and Unforeseen Corrective.

- **Cost of Downtime:** Loss of income due to Forced and Planned Unavailability is calculated by multiplying the lost electricity production during the downtime by the market price of electricity. These figures can be market-adjusted to suit specific analytical purposes.

- **Availability:** Actual availability refers to the percentage of time a plant is ready to produce electricity. Market Adjusted Availability (MAA) takes this a step further by excluding the periods when the electricity price is lower than the value of the water used to generate it. In such cases, the plant should not be producing, regardless of its readiness.

Figure 8 is an example of the visualization created using Microsoft Power Business Intelligence Software (Power BI) showing the proportion of each category of Maintenance expenditures in percentage from year 2021-2024.

3.4 The Creation of Synthesized Data

To develop supervised ML models capable of predicting hydropower plant efficiency, we needed a training dataset with labeled efficiency scores. However, the real-world dataset lacked this critical label. As such, we constructed a synthetic dataset grounded in actual operational data

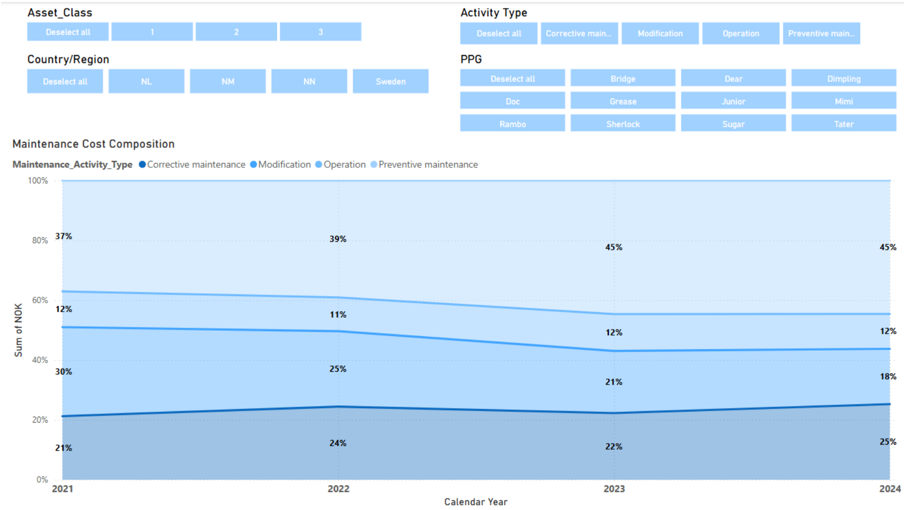


Figure 8: Maintenance Expenditure Composition.

Note: All identifiable features have been altered to preserve anonymity.

but designed to include a column named "Constructed Efficiency Factor", a known *efficiency factor* - our target variable.

3.4.1 Rationale for Synthetic Data

Real hydropower datasets typically lack explicit efficiency scores. Without this label, supervised learning models cannot be trained. They can only perform clustering or unsupervised grouping. By generating synthetic records with known efficiency scores, we enable **supervised learning**, allowing models to learn and predict plant performance more effectively.

3.4.2 Generation Process:

We adopted a rule-based generation approach, using averaged real-world data and deterministic scaling rules based on capacity bins and efficiency scores. The following are the steps we took in creating the synthetic data.

1. **Use data from 2021 and 2023 as seed.** These years were selected for their completeness and relevance. Averaging data across both years helps mitigate missing values and prevents data leakage between training and prediction phases. The dataset was first cleaned and filtered, after which numeric features were averaged over the two-year period.

2. **Group plants by installed capacity into bins.** To account for variability in plant size, power plants were grouped into bins based on installed capacity. We made sure that the synthetic data is generate with the similar Capacity Bin Distribution. The final results are shown in Table 1:

Capacity Bin	**Range (MW)**	**Seed Data (%)**	**Synthetic Data (%)**
1	1–150	82.22	82.55
2	151–299	9.62	9.39
3	300–449	3.70	4.00
4	450–599	2.22	2.01
5	600+	2.22	2.01

Table 1: Distribution of power plants by capacity bin in Seed and Synthetic datasets

3. **Calculate bin-level feature averages.** Within each bin, average values were computed for key features such as production volume, maintenance frequency, and availability. These bin-level averages served as baseline values for synthetic data generation.

Capacity Bin	Actual ATD %	PLN MAU ATD (MWh)	FRC MAU ATD (MWh)	Installed Capacity (MW)
1	94.67	1048.72	393.33	30.05
2	93.02	8254.60	1829.70	208.35
3	90.17	18921.63	5456.18	338.60
4	83.39	57327.66	6816.04	533.33
5	93.33	37219.84	9134.11	833.33

Table 2: Selected Metrics by Capacity Bin

4. **Generate synthetic records with P-values from 5–95%.** Each synthetic record was assigned a known *efficiency score*, expressed as a **P-value** ranging from **5% to 95% in 10% increments**. This score acts as the supervised learning label for model training.

5. **Scale features according to their efficiency relationship.** Features were scaled based on their relationship to efficiency as follows:

    - **Higher values indicate better performance** (e.g., actual availability, total production):
      Values were scaled proportionally to the P-value.

    - **Lower values indicate better performance** (e.g., maintenance downtime):
      Values were scaled using the inverse: $(100 - P)\% \times$ average.

    - **Neutral or ambiguous indicators**:
      Scaled using a simplified average-based factor for consistency.

6. **Add controlled noise/variation.** To simulate real-world fluctuations, we introduced a controlled random variation of $\pm 5\%$ into the synthetic data. This adds noise to prevent overfitting while preserving meaningful feature relationships required for model training.

7. **Validate distributions and relationships.** To validate the realism of the generated data, we compared the synthetic and real datasets using:

    - Feature distribution comparisons

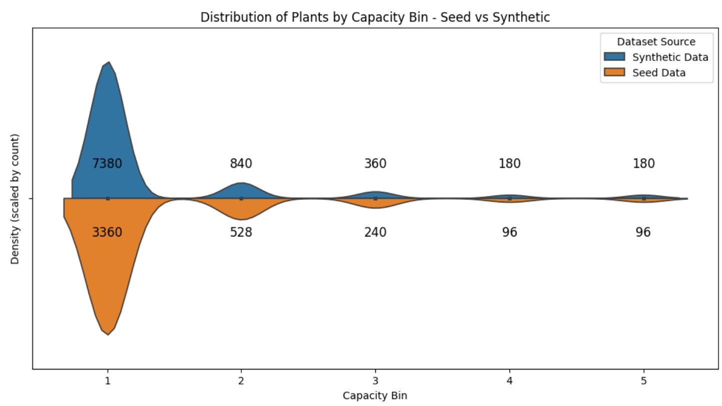


Figure 9: Distribution of Seed and Synthetic Data Across Capacity Bins

As shown in Figure 9 comparing the distribution based on Capacity_Bins found in "Seed" and "Synthesized" data are very similar. This suggest that the synthetic data is a valid substitute for the real data during training.

A comprehensive comparison of feature distributions using violin plots is provided in Figure 29 in the Appendix.

- **Comparing the pattern in Seed VS Synthetic Data**

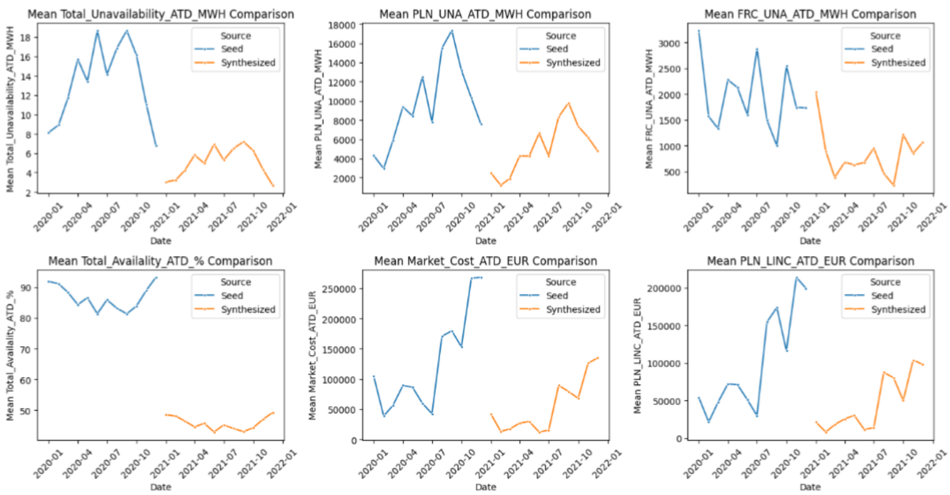


Figure 10: Seed VS Synthetic Data Pattern Comparison

Figure 10 shows that the synthetic data effectively captures the patterns present in the seed data, which serves as a proxy for the real data through averaging of the 2021 and 2023 datasets. The difference in height is due to the synthetic data being weighted by the constructed "Efficiency Factor," as described in Step 4 of the "Generation Process" section.

- **Correlation matrices** To ascertain the synthetic data accurately reflects the relationships present in the real-world dataset, we used correlation matrices to compare the feature inter-dependencies in both the "Seed" and "Synthetic" datasets.

  Figure 11 illustrate that the correlation patterns observed were highly consistent across the two, indicating that the same underlying relationships between selected features were preserved. This validation step is crucial, as it confirms that the model trained on synthetic data is likely to learn and generalize patterns that are representative of the real data, thereby enhancing its applicability and reliability in real-world scenarios.

The combined steps outlined above confirm that the synthetic data retains primary structures and statistic characteristics of the seed dataset. This is a strong indication that the ML models trained on the synthetic data are well-positioned to learn the latent patterns effectively. As a result, they are more likely to generalize well to real-world data and produce accurate predictions of the highly sought-after "Maintenance-based Efficiency Score."

3.4.3 Data Preprocessing and Cleaning

The process of converting compiled data into a structured format suitable for ML applications involves several preprocessing steps are carried out. A key task is plant name alignment and

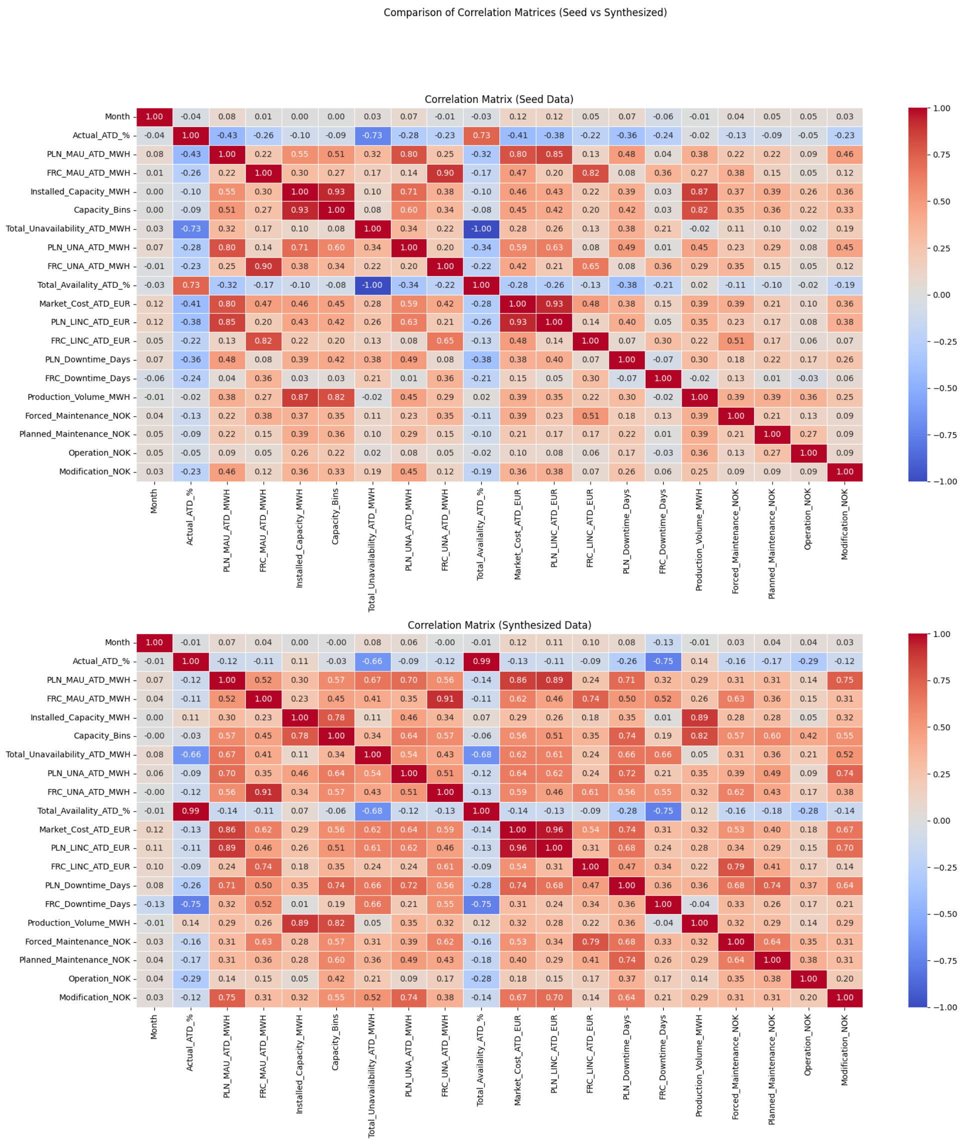


Figure 11: Correlation Heatmap

normalization, where unstructured maintenance logs and operational reports are processed to extract relevant information, including failure types, repair durations. Feature engineering is then applied to generate new variables that offer deeper insights, such as maintenance days per turbine and generator, downtime trends, and failure frequency.

To ensure consistency across the dataset, numerical values are standardized, mitigating discrepancies in measurement scales. These preprocessing steps are essential in refining the dataset, enhancing both model accuracy and interpretability, and ensuring that the information is structured in a way that allows for meaningful pattern detection and anomaly identification. The following steps were undertaken:

- **Cleaning:** The process of building the desired dataset started with basic data cleaning, such as dropping unnecessary columns and ensuring numerical columns are correctly formatted as integers or float. We standardized the naming scheme across the datasets to better identify the relevant columns and to make them easier to work with (e.g., removing things such as "[-]" and linebreaks such as "`\r\n`").

- **Standardizing:** We standardized the "Plant_ID" and "Plant_name" to enable the merge of data from different datasets and queries.

- **Transforming:** Several data transformations were performed to tailor the dataset for our specific use. Below are some of the most notable ones:

  - **Adjust the DateTime data:** We also transformed all date columns from a DateTime format to separate Year and Month columns as Integers. This was necessary because some data was only available on a monthly scale. Additionally, since both monthly and yearly data were present across all three time series datasets, syncing the time intervals was crucial for proper merging.

  - **Transpose Maintenance Logs:** The maintenance logs was in a wide format while the others were in a long format, therefore we melted the maintenance logs transforming it into a long format.

  - **Capture the number of downtime days:** The record gives us the start and end time (year-month-date-time) of all downtime incidences, we extracted the number of downtime days across the months, and derived at two aggregated columns: "Planned downtime Days and Forced downtime Days."

- **Handling Missing Values:** Missing values were found only in the "JobTypes" metadata column, which we replaced with 0 to indicate a missing job type. Impute missing data does not make sense for the data we have, as adding down time, production or availability changes the facts of the data.

- **Handling Data Inconsistencies:** The main challenge was the use of different identification methods for power plants. For example, the SAC Availability query used a different system than other sources, making direct matching impossible. Inconsistent plant names further hindered automated matching, requiring manual comparisons. A list of discrepancies and proposed standardized `Plant_IDs` was submitted to Statkraft for review. After approval, we obtained a unified, standardized dataset— the `Plant_Info` dataset—containing all hydropower plants and their associated information.

- **Final dataset for modeling:** The final dataset after processing consists of 24 columns as shown in Table 3 bellow.

3.5 Machine Learning Models

This section explores two groups of ML models for performance prediction and anomaly detection. The first group includes tree-based methods like XGBoost and Random Forest, known for their robustness and interpretability. The second group focuses on neural network-based approaches, such as AutoEncoders and MLP, ideal for capturing complex, nonlinear relationships in high-dimensional maintenance data.

3.5.1 Machine Learning Pipeline

To develop the MBE-score and apply it effectively to real-world hydropower plant data, we followed a structured ML pipeline comprising several critical stages:

1. **Data Collection and Preprocessing:** The process began with collecting data from multiple sources, which was cleaned, filtered, and standardized to address missing values and inconsistencies. Imputation techniques were applied to ensure compatibility. Due to the absence of labeled efficiency scores, synthetic data with known outputs was generated.

Table 3: Final dataset for modeling

Field	Description
Plant_ID	Unique identifier (4 chars) for the power plant
Plant_Name	Name of the power plant
Year	Reporting year
Month	Reporting month
Capacity_Bins	Capacity classification based on installed capacity
Actual_ATD_%	Market adjusted availability percentage
PLN_MAU_ATD_MWH	Market adjusted Planned Unavailability in MWh
FRC_MAU_ATD_MWH	Market adjusted Forced Unavailability in MWh
Installed_Capacity_MW	Installed capacity in MW
Total_Unavailability_ATD_MWH	Total unavailability in MWh
PLN_UNA_ATD_MWH	Planned unavailability in MWh
FRC_UNA_ATD_MWH	Forced unavailability in MWh
Total_Availality_ATD_%	Total availability percentage
Market_Cost_ATD_EUR	Market cost in EUR
PLN_LINC_ATD_EUR	Lost income during Planned maintenance in EUR
FRC_LINC_ATD_EUR	Lost income during forced maintenance in EUR
PLN_Downtime_Days	Planned downtime in days
FRC_Downtime_Days	Forced downtime in days
Production_Volume_MWH	Volume of electricity produced in MWh
Forced_Maintenance_NOK	Cost of forced maintenance in NOK
Planned_Maintenance_NOK	Cost of planned maintenance in NOK
Operation_NOK	Maintenance related operational cost in NOK
Modification_NOK	Cost of plant modifications in NOK
Constructed_Efficiency	Efficiency Factor used in constructing synthetic data (proxy for target feature)

These datasets were designed to mirror the real data's structure and statistical properties, with correlation matrices and feature distributions compared to ensure generalization to real-world scenarios.

2. **Model Training:** The models were trained using a semi-synthesized dataset, which combines elements of synthetic data (with a known labels "Constructed Efficiency Factor") and real-world data structures. This enabled us to guide the learning process while preserving realistic input features. We trained four distinct regression models:

   - XGBoost Regressor
   - Random Forest Regressor
   - Autoencoder + Regressor
   - MLP

3. **Hyperparameter Tuning:** Each model underwent hyperparameter optimization using RandomizedSearchCV, with 50 randomized trials per model. This allowed us to efficiently explore the hyperparameter space and identify configurations that offered the best performance.

4. **Model Validation and Evaluation:** K-fold cross-validation was employed, dividing the dataset into k equally sized subsets. In each iteration, the model was trained on $k-1$ subsets and evaluated on the remaining one. This process was repeated k times, ensuring

that each subset served as the test set once. Averaging the results across all folds reduced the impact of any single data split and provided more reliable performance estimates.

Model performance was evaluated using R^2, RMSE, and MAE. During the hyperparameter grid search, RMSE served as the primary criterion within the training loop for selecting the best parameters. For the final model selection, all three metrics were considered collectively.

5. **Imputation on Real data for predicting efficiency:** To address missing values in the real dataset and ensure compatibility with ML models, we applied a **two-stage imputation** strategy. In the **first round**, missing values were imputed using the average of the corresponding feature within the same year, preserving year-specific characteristics. In the **second round**, any remaining missing values were filled using the average across all available years for that feature. This approach helped maintain temporal consistency and avoided mixing data between different plants, ensuring the integrity of plant-specific patterns.

   Most plants located in Sweden that lacked sufficient maintenance expenditure data to allow for meaningful imputation were excluded from the analysis—a total of 49 plants. This step was necessary to ensure data compatibility and reliability before using the dataset in the model during the prediction phase. As a result, predictions were ultimately made on a total of only 90 unique plants.

6. **Application to Real Data:** Once trained and validated, the optimized models were applied to real-world data to predict the MBE scores. The objective was to quantify the impact of maintenance activities on plant performance, providing Statkraft with a valuable tool for further analysis and the identification of hidden inefficiencies. These insights can serve as a foundation for refining maintenance strategies and improving overall operational efficiency.

3.6 Tools Used

During our bachelor project, we utilized a range of pre-approved software tools, as Statkraft maintains strict controls over what is permitted within its network. The following is a list of the tools we employed, along with a brief description of their respective purposes in supporting data management and achieving our project objectives.

- **Microsoft Teams** is a collaboration platform that integrates chat, video conferencing, file sharing, and calendar scheduling in a single space. It enables real-time communication and effective coordination, especially across distributed teams.

- **SAP Analysis for Microsoft Office** is an Excel-based tool that allows users to access, analyze, and report on data from SAP Business Warehouse (BW) or SAP HANA (High-Performance Analytic Appliance) systems using the familiar spreadsheet interface.

  We used it to extract data from SAP Statkraft systems and save in .csv format for further processing.

- **Visual Studio Code (VS Code)** is a source-code editor developed by Microsoft. It supports various programming languages, and provides a wide range of features, including integrated version control.

  VS Code was used as the primary Integrated Development Environment (IDE) through the project for writing Python code, performing data processing, and developing ML models.

- **DataWrangler:** This is a VS Code extension designed to simplify data exploration, cleaning, and visualization. The tool was created with the aim to make Exploratory Data Analysis (EDA) faster and more efficient. It offers user friendly, spreadsheet interface for working with tabular data formats.

- **Microsoft Power BI** is an analytics and data visualization platform designed for transforming complex datasets into clear, visual insights. It offers a user-friendly interface with strong data modeling capabilities and enables users to combine data from various sources and present it with visual reports.

  For this project Power BI was used to explore patterns in maintenance, production and availability data by developing interactive dashboards.

- **GitLab** is a source code hosting platform used for version control, that provides a structured environment for collaborative work. It combines tools for storing files, tracking progress, and keeping complex projects organized.

  In this project, GitLab was used as shared workspace for project code, supporting files and related materials. Version control system allowed team to always stay up to date with the current state of the project, track progress, and coordinate contributions.

4 The Solution/Product

At the start of the project, we explored various ML models to predict hydropower plant efficiency, including classification and complex architectures. As the use of synthetic data enabled continuous efficiency prediction, we reframed the task as a regression problem. We ultimately selected four non-linear models for comparison—MLP, XGBoost, Random Forest, and an Autoencoder-based regressor—representing both deep learning and ensemble approaches capable of capturing complex, nonlinear relationships.

4.1 Candidate Models

This section presents the ML models used to predict MBE scores: XGBoost Regressor for its boosting performance, Random Forest Regressor for robustness and interpretability, an AutoEncoder with a Regressor for feature compression, and a MLP Regressor for capturing nonlinear relationships.

4.1.1 XGBoost Regressor

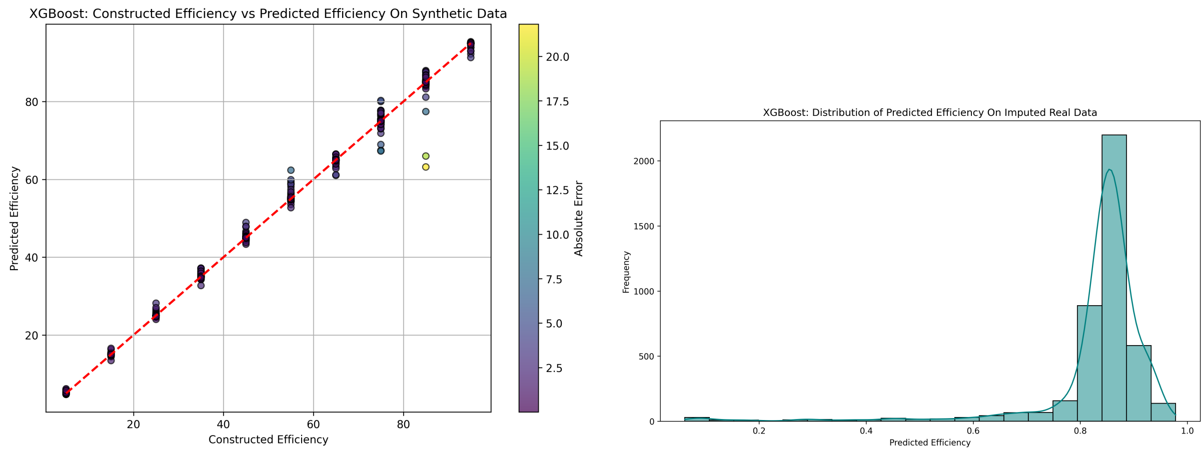
The XGBoost pipeline begins with final preprocessing steps, including dropping categorical data, defining the target, and performing an 80/20 train-test split. An XGBRegressor (1,000 trees, squared-error objective, RMSE evaluation) is instantiated using the xgboost library and tuned via RandomizedSearchCV (50 iterations, 10-fold CV, scoring = negative RMSE). The hyperparameter search space is shown in Table 4. A boosted ensemble of decision trees forms the regression head, outputting efficiency scores on a 0–100 scale. The final model is then applied to the imputed real-world dataset for prediction and visualization.

Table 4: XGBoost Hyper Parameters

Parameter	Search space		
learning_rate	0.01	**0.05**	0.1
max_depth	3	5	**7**
min_child_weight	1	**5**	
gamma	**0**	1	
subsample	**0.7**	1.0	
colsample_bytree	**0.7**	1.0	
booster	**gbtree**	dart	

Notes: The best performing parameters are in **bold**.

XGBoost Training Results The best hyperparameters found during the experiment and are shown in Table 4 above. On the 20 % hold-out test set, this model achieved $R^2 = 0.998960$, $RMSE = 0.941573$ and $MAE = 0.195090$. Table 8 compares the metrics with other candidate models. The regressor outputs efficiency directly on a 0–100 scale, so RMSE 0.94 translates to 0.94 percentage-points.



(a) Synthetic: CE vs. PE

(b) Imputed real data: PE distribution

Figure 12: XGBoost Test set performance on synthetic (left) and Predictions on real (right).

Figure 12a plots predicted vs. constructed efficiency on the synthetic data (points colored by absolute error). The tight clustering around the diagonal shows that XGBoost accurately captures the underlying signal with minimal deviation, corroborating the low error metrics.

Applying XGBoost to Imputed Real-World Data The predicted efficiencies on the real-world data form a highly leptokurtic distribution centered around 84 %, with a long left tail and a truncated right tail bounded at 1, as shown in Figure 12b. Figure 23 in the appendix breaks this down further by capacity bin. Statistical metrics reinforce this pattern: Fisher's kurtosis is 19.8994, skewness is -4.0892 and the Mode is 0.8428, confirming a sharp peak and negative skew. The comparison of these statistical characteristics with other models is shown in Table 9.

The model draws most of its signal from *Actual_ATD_%*, *Total_Availability_ATD_%* while *Total_Unavailability_ATD_MWH* carries a strong negative importance. All other inputs contribute weak-to-moderate negative gains; *Installed_Capacity_MW*, and the cost oriented features (*Planned_Maintenance_NOK*, *Forced_Maintenance_NOK*, *Operational_NOK*, and *Modi-*

fication_NOK) are close to neutral. However The feature importance varies greatly between capacity bins as shown in Figure 13.

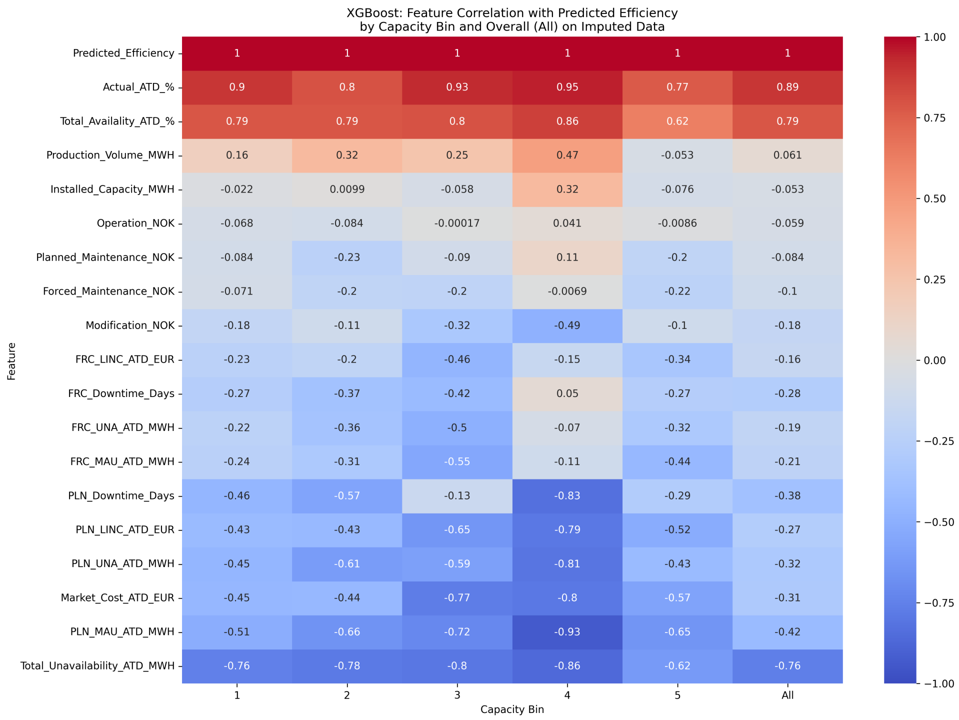


Figure 13: XGBoost: Feature Correlation Comparison Across Capacity Bin.

4.1.2 Random Forest Regressor

The Random Forest pipeline mirrors the XGBoost workflow: after final preprocessing (dropping categorical data, defining the target, and an 80/20 train-test split), we instantiate a `Random Forest Regressor` from scikit-learn and tune it using `RandomizedSearchCV` (50 iterations, 10-fold CV, scoring = negative RMSE). The optimal model uses 978 trees with squared error as the criterion. The hyperparameter search space is shown in Table 5. The model predicts efficiency as the mean output of all trees, which is then applied to the imputed real-world dataset for visualization.

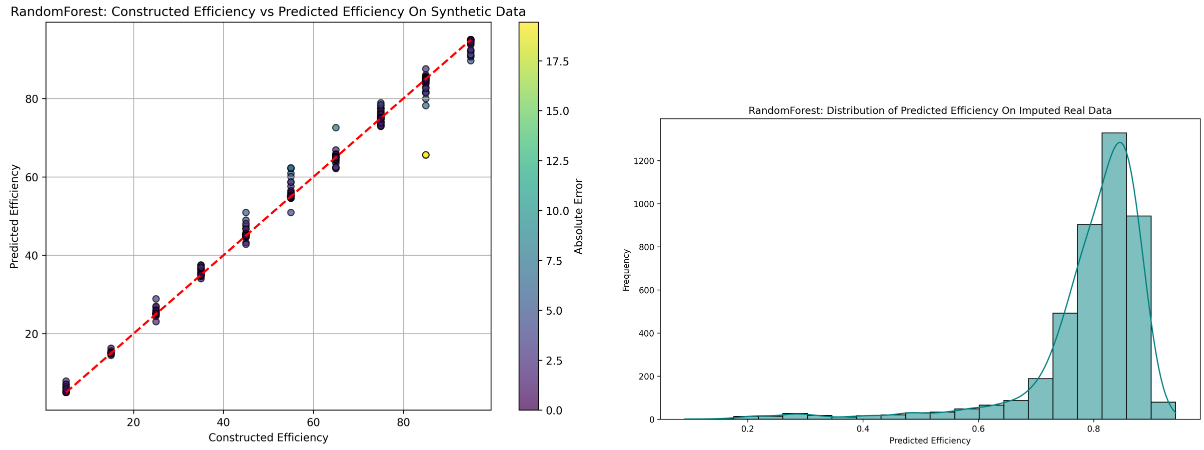
Table 5: Random Forest Hyper-Parameters

Parameter	**Search space**			
n_estimators	100	**978**	1 000	
max_depth	10	**17**	30	
min_samples_split	2	**3**	10	
min_samples_leaf	**1**	...	3	
max_features	**sqrt**	log2		
bootstrap	**True**	False		
criterion	**squared_error**	absolute_error		
max_samples	None	0.50	0.75	**1.00**

Notes: The best performing parameters are in **bold**.

Random Forest Training Results The optimal configuration during the experiment and are shown in Table 5 above. On the 20 % hold-out test set this model achieved $R^2 = 0.998927$, $RMSE = 0.956387$ and $MAE = 0.190836$. Table 8 compares the metrics with other candidate models. The forest predicts on a 0–100 scale, meaning RMSE 0.95 equals about 95 percentage-points of error.

Figure 14a shows predicted vs. constructed efficiency on synthetic data; The tight clustering along the 45° line confirms that the Random Forest model captures the embedded relationships with minimal error.



(a) Synthetic: CE vs. PE

(b) Imputed real data: PE distribution

Figure 14: Random Forest test-set performance on synthetic (left) and predictions on real (right).

Applying Random Forest to Imputed Real-World Data Figure 14b show predicted efficiencies forming a leptokurtic distribution centered round 81 %, with pronounced left skew (Fisher kurtosis = 10.4199, skewness = -2.8888, Mode = 0.8183), echoing the pattern seen in XGBoost. The comparison of these statistical characteristics with other models is shown in Table 9. Figure 24 in the appendix splits this by capacity bin.

The forest's feature importance mirrors the XGB model result: Actual_ATD_% and Total_Availability_ATD_% carry the strongest positive weight, Total_Unavailability_ATD_MWH contributes a strong negative signal, while Installed_Capacity_MW and cost features remain effectively neutral. Importance shifts significantly between capacity bins as shown in Figure 15.

4.1.3 AutoEncoder+Regressor

This two-stage pipeline is built as a **TensorFlow / Keras** stack. We first train an unsupervised Keras autoencoder: input → 128 → 64 → **32-node bottleneck** → 64 → 128 → output, using MSE loss and early stopping. After convergence we freeze TensorFlow's encoder and drop the decoder; the 32-dimensional embeddings produced by the encoder becomes features for a shallow feed-forward network whose single-neuron output layer serves as the model's regression head.

Following the usual preprocessing (drop identifiers, `MinMaxScaler`, 80% train / 20% test), we tune the regressor's hyper-parameters with scikit-learn's `RandomizedSearchCV` (50 trials, 10-fold CV, scoring =;–RMSE). The search grid appears in Table 6. Finally, the final encoder + regressor pair is applied to the imputed real-world dataset and its predictions visualized.

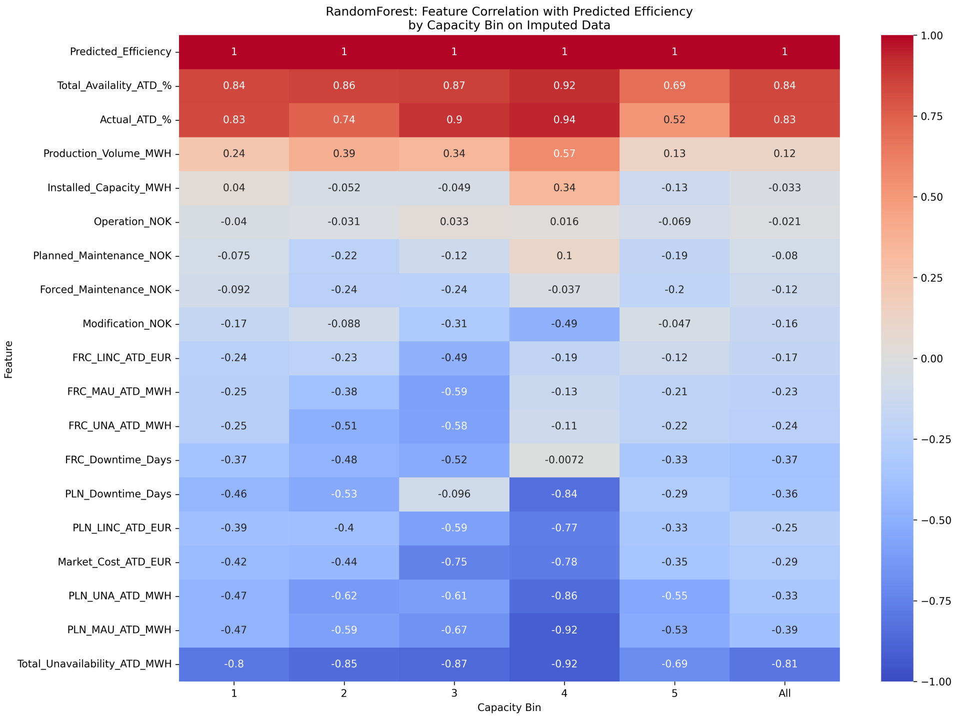


Figure 15: Random Forest: Feature Correlation Comparison Across Capacity Bin.

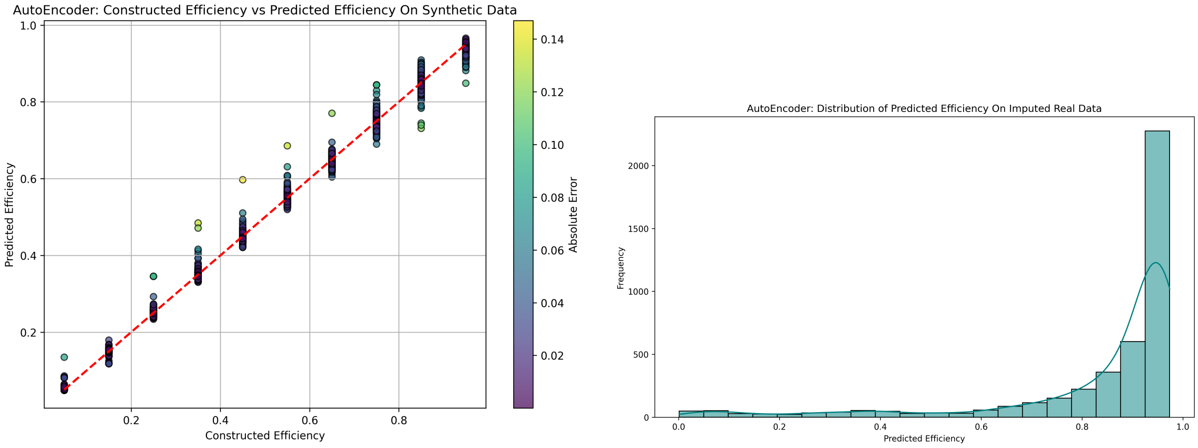
Table 6: AutoEncoder + Regressor Hyper-Parameters

Parameter	Search space		
hidden_layer_sizes	(64, 32)	(128, 64)	**(256, 128)**
activation	**relu**	tanh	
optimizer	**adam**	rmsprop	sgd
learning_rate	0.01	**0.001**	0.0001
epochs	50	**560**	900
batch_size	16	**41**	64

Notes: The best performing parameters are in **bold**.

AutoEncoder + Regressor Training Results As shown in Table 6, randomized search selected *rmsprop* (*learning_rate*=0.001) with hidden layers (128, 64), *batch_size*=43 and *epochs*=522 as the best parameters. On the 20 % hold-out test set the model reached $R^2 = 0.995717$, $RMSE = 0.019111$ and $MAE = 0.012051$. Table 8 compares the metrics with other candidate models. A the last layer uses sigmoid activation which bounds output predictions to [0, 1]; errors are therefore on a 0–1 scale—RMSE 0.019 translates to 1.9 percentage-points, for comparison with tree based models we scale RMSE and MAE with 100 for the AutoEncorder.

Figure 16a shows a slight reverse S-shape; Low-efficiency plants sit just above the 45° line and high-efficiency plants just below it, indicating a mild compression of the range. On real data the predicted-efficiency histogram (Figure 16b) show a distribution closer to an exponential drop-off than the sharp leptokurtic spike seen in the tree based models. It clusters around 95 % but with a more pronounced left skew. This is quantified by Fisher kurtosis at 4.7763, skew at -2.3234 and a Mode at 0.9565. The comparison of these statistical characteristics with other models is shown in Table 9.



(a) Synthetic: CE vs. PE (b) Imputed real data: PE distribution

Figure 16: AutoEncoder + Regressor test-set performance on synthetic (left) and predictions on real (right).

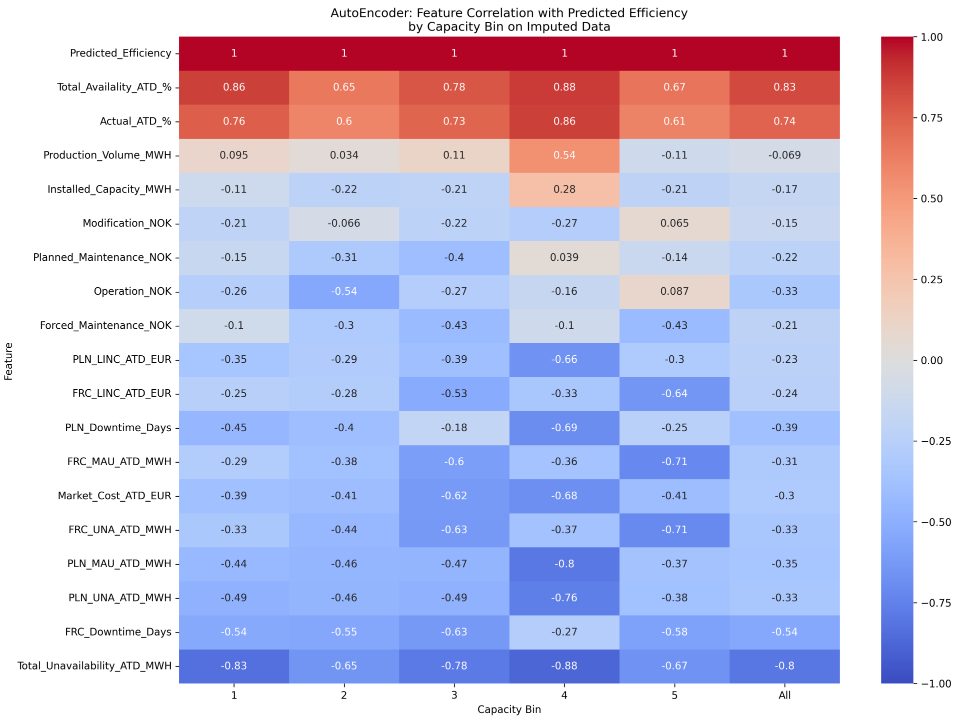


Figure 17: AutoEncoder + Regressor: Feature Correlation Comparison Across Capacity Bin.

The model singles out *Actual_ATD_%* and *Total_Availability_ATD_%* as the strongest positive drivers, with *Total_Unavailability_ATD_MWH* remaining negative. In contrast as seen in Figure 17 the Production_Volume_MWH and Installed_Capasity_MW are largely ignored by the model. Notably the cost related features have a slight negative importance here.

4.1.4 Multi-Layer Perceptron Regressor

This is a fully connected TensorFlow / Keras network wrapped with scikeras so it plugs into scikit-learn. A Sequential graph takes a p-dimensional input (in this case X_train.shape[1]), passes it through a stack of Dense layers whose widths are drawn from the tuple layer_sizes.

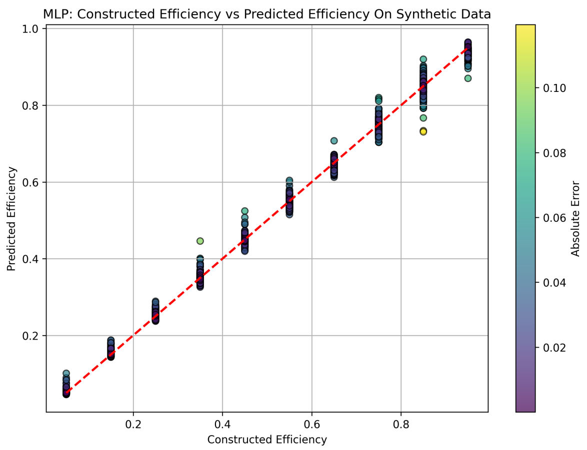
Each layer uses either ReLU or tanh and can be followed by an optional Dropout(dropout_rate) block when dropout rate isn't 0. A single-neuron output layer with sigmoid activation returns a 0–1 bounded efficiency estimate—this neuron is the network's regression head. The model is compiled with MSE loss and selects an optimizer at a sampled learning-rate. After preprocessing (identifier drop, `MinMaxScaler`, 80 % train / 20% test) we tune these hyper-parameters with `RandomizedSearchCV` (50 trials, 10-fold CV, scoring = –RMSE); the search grid is summarized in Table 7. The final MLP model is then applied to the imputed real-world dataset.

Table 7: MLP Hyper-Parameters

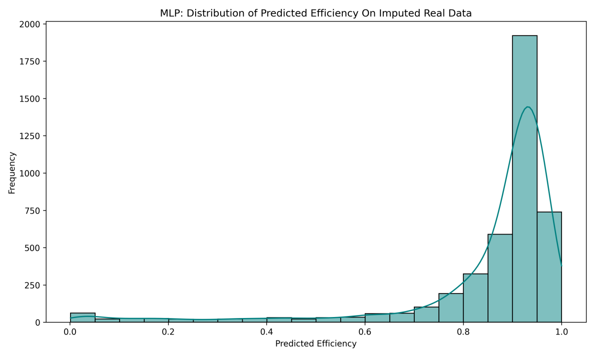
Parameter	Search space		
layer_sizes	...	**(256, 128, 64, 32)**	...
activation	**relu**	tanh	
optimizer	**adam**	rmsprop	
learning_rate	**0.001**	0.0005	0.0001
dropout_rate	**0.0**	0.2	
batch_size	**64**	32	
epochs	300 (early-stop)		

Notes: The best performing parameters are in **bold**.

MLP Training Results The optimal configuration found during the experiment and are shown in Table 7 above. On the 20 % hold-out test set the network achieved $R^2 = 0.996584$, $RMSE = 0.017069$ and $MAE = 0.012048$. Table 8 compares the metrics with other candidate models. The final layer uses activation sigmoid which keeps predictions bounded to [0, 1]; RMSE 0.024 translates to 2.4 percentage-points on the 0–100 scale, for comparison with tree based models we scale RMSE and MAE with 100 for the MLP model.



(a) Synthetic: CE vs. PE



(b) Imputed real data: PE distribution

Figure 18: MLP test-set performance on synthetic (left) and predictions on real (right).

Figure 18a reveals a gentle upward bias across the synthetic range-predictions, averaging slightly above the diagonal until roughly 80 % efficiency, after which they dip below, signaling minor over- then under-estimation. On the real world data Figure 18b shows the prediction distribution peaks near 92 % with a heavy left skew. The distribution tapers off fast above the peak. This is quantified by a Fisher kurtosis of 8.5237, a Skewness of -2.9313, and a mode of 0.9241. The comparison of these statistical characteristics with other models is shown in Table 9.

Figure 19 again flags *Actual_ATD_%* and *Total_Availability_ATD_%* as dominant positives, however, Total_Unavailability_ATD_MWH is almost rivaled by PLN_Downtime_Days in negative importance, and FRC_Downtime_Days are also heavily weighted. In addition. No columns seem to be completely unused by the model but Installed_Capacity_MW and Production_Volumn_MWH are still very weak.

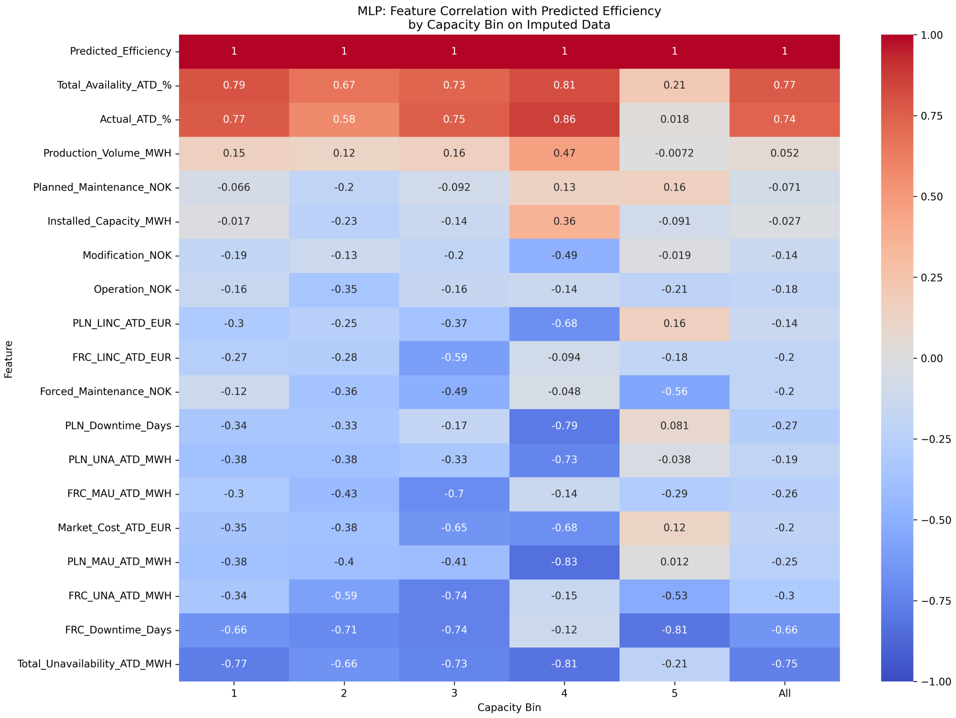


Figure 19: MLP: Feature Correlation Comparison Across Capacity Bin.

Table 8: Evaluation Metrics for Each Model on the Test Set

Model	R^2	RMSE	MAE
XGBoost Regressor	0.998960	0.941573	0.195090
Random Forest Regressor	0.998927	0.956387	0.190836
AutoEncoder + Regressor	0.995717	1.9111	1.2051
MLP Regressor	0.996584	1.7069	1.2048

Notes: AutoEncoder and MLP model RMSE and MAE are scaled by 100 be be comparable with the tree models.

The result metrics for each model previously described in its respective model sections are now shown in Table 8 for ease of across model comparison. In-depth discussion is done in the next section, "Selecting the Best Model".

Table 9: kurtosis and Skewness for Each Model on the Test Real Imputed Data

Model	Fisher	Pearson	Skewness	Mode
XGBoost Regressor	19.8994	22.8994	-4.0892	0.8428
Random Forest Regressor	10.4199	13.4199	-2.8888	0.8183
AutoEncoder + Regressor	4.7763	7.7763	-2.3234	0.9565
MLP Regressor	8.5237	11.5237	-2.9313	0.9241

The previously discussed statistical characteristics (Kurtosis, Skewness, and Mode) in each section of the model are compiled in Table 9. The XGBoost Regressor exhibits the highest kurtosis (Fisher: 19.90) and the most pronounced negative skewness (-4.09), indicating a distribution with heavy tails and a strong leftward asymmetry. The Random Forest Regressor also shows elevated kurtosis (10.41) and skewness (-2.89), though to a lesser extent. The AutoEncoder + Regressor and MLP Regressor demonstrate more moderate values, with the MLP showing the lowest kurtosis (3.28) and least skewed distribution (-2.12). The Mode metric remains relatively stable across models, ranging from 0.8183 to 0.9415, suggesting limited variability in peak distribution values.

4.2 Selecting the Best Model

This section outlines the process used to evaluate and select the most suitable model architecture for predicting the MBE-score.

Performance Comparison Across Models

Following the model-level experiment described in Section 4.1, 50 hyperparameter sets were evaluated using 10-fold cross-validation, with training and testing losses recorded. Figure 16 compares the losses on synthetic training data, sorted by training loss, helping to identify signs of overfitting.

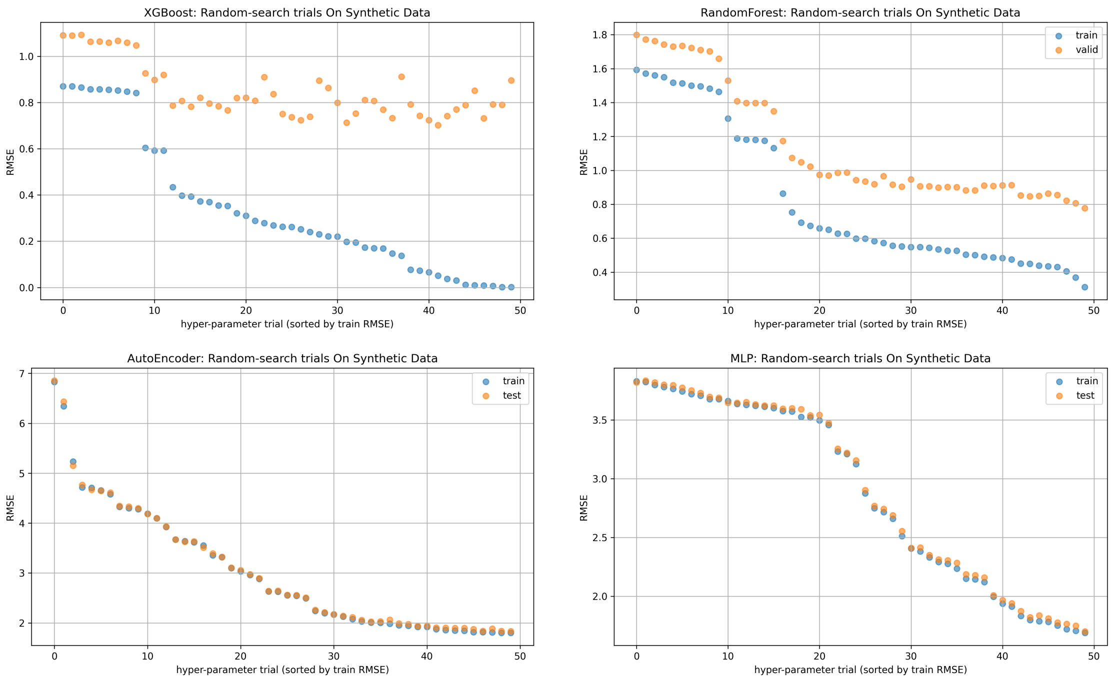


Figure 20: Hyperparameter search space results comparison
Note: The AutoEncoder and MLP model results are scaled for comparison.

Figure 20 shows a clear gap between training and testing RMSE in 10-fold cross-validation for the tree-based models. XGBoost displays notable overfitting, with test RMSE flattening while training error drops. Random Forest also overfits, though test error improves slightly.

In contrast, the deep learning models—AutoEncoder and MLP—maintain nearly identical training and testing RMSE across all folds, indicating minimal overfitting and strong generalization potential for real-world data.

Next, each model is trained on the synthetic data using the best hyperparameters from the previous step, with evaluation metrics—R^2, RMSE, and MAE—recorded for comparison. Figure 21 visualizes the results.

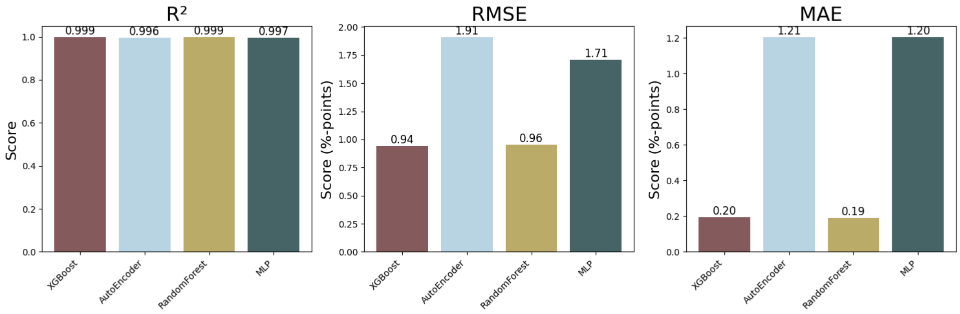


Figure 21: Evaluation Metric Comparison Across Candidate Models.
Note: The AutoEncoder and MLP model results are scaled for comparison.

The metrics shown in Figure 21 offer key insights into model accuracy and reliability. Lower RMSE and MAE values indicate smaller average prediction errors, with RMSE being particularly sensitive to large deviations—useful when such errors carry operational risk. Ideally, a strong model combines a high R^2 (close to 1) with low RMSE and MAE (close to 0).

In this evaluation, the deep learning models showed better generalization, with smaller gaps between training and testing performance. However, they underperformed compared to tree-based models in predictive accuracy. The tree-based models, while more accurate, showed greater variance in generalization.

Based on both evaluation metrics and generalization ability, **Random Forest emerges as the best-performing model**.

4.3 Prediction based on the selected model.

This section introduces the application of the selected predictive model—Random Forest—to real-world data in order to generate MBE-score for Statkraft's hydropower plants.

Building on the synthetic data experiments and model evaluation presented in Section 4.2, the trained Random Forest model is applied to imputed real data from 2021 to 2024 to generate efficiency scores that represent the relative operational performance of each plant. These scores are then integrated with additional plant-level information in PowerBI to produce the visualization shown in Figure 22.

5 Discussion

This section evaluates the project outcomes, starting with a review of the final deliverables and their alignment with Statkraft's goals. It then addresses practical challenges for real-world deployment and concludes with recommendations for improving accuracy, scalability, and system integration.

5.1 Reflections on Deliverables

By the end of the four-month project, several key outputs were developed and delivered to Statkraft. The following is a summary of the main deliverables and possible use-cases for them:

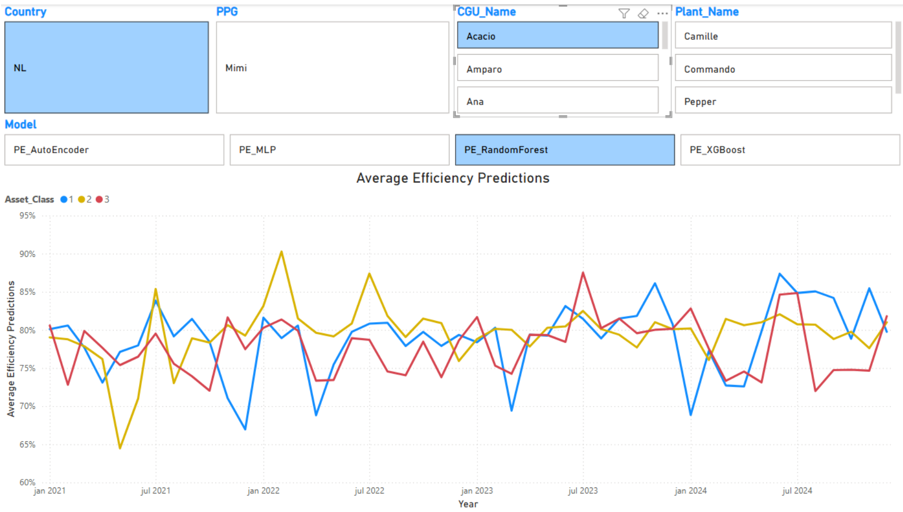


Figure 22: Example of Distribution of Predicted Efficiency

Note: All identifiable features have been altered to preserve anonymity.

- **Power BI: Historical Maintenance Analysis** Completed at the end of March and handed over to Statkraft, this tool allows for in-depth analysis with "Asset Class" as the primary comparison point across multiple hierarchical levels—from country/region to individual plants. The interactive visualizations support analysis across various data sources, enabling comparisons of key metrics like production levels, maintenance costs per unit of production, maintenance expenditure composition, and downtime. Analyses can be done for individual years (2021-2024) or over selected time periods, including time series analysis. Additionally, market-adjusted planned and forced downtime and associated income losses are incorporated.

  Beyond static reporting, the Power BI dashboard serves as a dynamic decision-support tool. It helps identify cost drivers, detect trends, and explore correlations between operational variables and financial outcomes. This supports data-driven maintenance planning and decision-making. By allowing domain experts and managers to interact with the data without technical expertise, the tool enables Statkraft to take a more strategic role in operational planning, aligning data insights with long-term business objectives and optimizing performance.

- **Machine Learning Model Code for Efficiency Prediction**
  A complete package was delivered, including detailed environment setup instructions and Jupyter notebooks containing the full model architectures with thorough annotations, along with the notebook used for synthetic data generation. The models produce both time-series outputs for each power plant and a single aggregate efficiency score, offering flexible perspectives for performance assessment.

  This deliverable provides Statkraft with a robust foundation for continued development. The codebase can be used to further refine the synthetic data generation process, adjust or expand the model architectures, and adapt the pipeline for other datasets or prediction targets.

  In terms of application, the models can support early detection of performance degradation, benchmarking of plant efficiency across the plants, and prioritization of maintenance efforts based on predicted outcomes. Over time, this could lead to increased operational efficiency, reduced maintenance costs, and improved asset management. By having full access to transparent, modifiable code, Statkraft is well-positioned to integrate predictive analytics into their broader digital strategy.

- **Power BI: Efficiency Score Benchmarking Tool**
  This deliverable builds on the output from the ML model to visualize and categorize predicted efficiency scores across various business areas. It allows users to drill down from a high-level country or regional overview to the level of individual power plants, enabling detailed and intuitive performance analysis.

  The tool serves as a powerful resource for identifying plants that deviate from expected efficiency levels or fall outside the normal performance range. It can support early detection of underperformance or emerging operational issues. Simultaneously, it can highlight high-performing plants, making it easier to benchmark efficiency and identify operational best practices. These insights can inform possible targeted interventions, guide predictive maintenance strategies, and support strategic resource allocation.

  Beyond day-to-day monitoring, the benchmarking tool provides Statkraft with a data-driven framework for continuous improvement. By enabling cross-site comparisons and sharing successful strategies across locations, the organization can foster a culture of performance excellence. It also supports long-term strategic planning by aligning maintenance and operational decisions with predictive insights, thereby increasing efficiency, reducing downtime, and strengthening overall asset management capabilities.

In conclusion, this project demonstrates that ML models such as Random Forest can effectively model hydropower plant efficiency using historical operational data. While results are promising, future improvements should focus on expanding data diversity, automating retraining pipelines, and aligning model outputs with business workflows to maximize practical impact.

5.2 Practical Considerations for Deployment

The ML solution for MBE has shown promising results when tested on synthetic datasets. However, several important factors must be considered before moving toward deployment. Below are some key considerations that need to be addressed to ensure successful operational integration.

Due to time and resource constraints, we were unable to perform an exhaustive hyperparameter search for each model. This means that some models may not have been optimized to their full potential. This is also likely observed in Figure 20, where the test RMSE is clearly continuing to decent.

While the dataset was cleaned and preprocessed with care, the final feature set may not have captured all the important operational nuances of hydropower plants. For example, time-series behavior such as seasonal changes or historical underperformance was not accounted for in this static dataset.

The presence of negative values in columns such as *Actual_ATD_%*, *Total_Availability_ATD_%*, *PLN_LINC_ATD_EUR* and, *FRC_LINC_ATD_EUR* may distort the true influence of these features when predicting the Efficiency Score. Such anomalies can potentially dampen or obscure their actual impact on model performance. This issue was raised with the company and will require restructuring at a higher data governance or system level to ensure consistency and accuracy moving forward.
The ML models used in this project are trained exclusively on synthetic data. As such, the accuracy and reliability of their predictions are directly tied to the quality and realism of that

data. These models are designed to detect patterns present in the synthetic dataset. If key operational characteristics or anomalies are missing or poorly represented, the model will be unable to recognize them in practice.

Consequently, any decision-making based on these predictions must be approached with caution. Automated actions should not be taken without human oversight, as the models may miss or misinterpret real-world scenarios that differ from those simulated in the synthetic data. To ensure meaningful and actionable results, it is essential that the synthetic data reflect the full range of operational conditions, performance issues, and contextual nuances found in actual hydropower operations. Without this alignment, there is a risk of basing strategic decisions on incomplete or misleading outputs.

The model can be deployed through scheduled batch processes, such as weekly or monthly evaluations. This approach enables decision-makers to identify performance trends, deviations, or anomalies over time, supporting more proactive maintenance planning and informed resource allocation. It also reduces the need for continuous, real-time prediction, making integration into existing workflows more practical and cost-effective.

All things considered, even a moderately accurate predictive model can deliver significant business value—particularly when used for benchmarking, performance analysis, and identifying underperforming assets. Our Random Forest model is well-suited for these purposes, demonstrating strong predictive accuracy. It could be effectively integrated into a periodic reporting or analysis pipeline, where predicted efficiency scores are compared against actual operational data to verify their plausibility and ensure the model's outputs align with real-world performance. This would support long-term performance monitoring, enhance maintenance planning, and contribute to data-driven decision-making at scale.

5.3 Future Work Suggestions

To enhance the robustness, accuracy, and practical applicability of the proposed model, several avenues for future work are recommended. These suggestions aim to deepen the integration of domain-specific knowledge, improve real-time capabilities, and refine the quality of both real and synthetic data inputs, ultimately driving more reliable and insightful efficiency predictions.

- More features, specifically technical details about a plant, such as Number of Turbine and Generator, Bering types, sensor types, oil types used, And other data such as environmental elements(e.g, water hardness, silt amount going through the turbine, head(water height), water volume through the year), weather patterns over time.

- Integration with SCADA for real time prediction and warning capabilities. Such integration would enable real-time prediction and early warning capabilities. By leveraging operational sensor data—such as temperature, vibration, flow rate, pressure, and output levels—the model could continuously monitor the health and performance of critical equipment. This real-time data stream would allow the model to detect anomalies, predict efficiency drops, and flag potential issues before they escalate, supporting a shift from reactive to predictive maintenance strategies. Furthermore, incorporating live SCADA data would enhance the model's accuracy over time by providing a more dynamic and granular view of plant operations, improving responsiveness and reducing unplanned downtime.

- Improve the complexity of the synthetic data to better capture real-world patterns and operational nuances, thereby enhancing the accuracy of the efficiency predictions. This includes aligning the synthetic data more closely with actual historical data to ensure that the model learns from realistic scenarios and produces outputs that are relevant and actionable in real-world contexts.

- Further analysis could involve examining the predicted efficiency scores in relation to factors such as turbine type or the age of equipment, whether the turbines, generators, or the overall facility. This would help determine whether specific turbine types or aging infrastructure have a measurable impact on maintenance needs and, consequently, influence the efficiency score of the plant.

6 Conclusion

This bachelor project has provided us with invaluable hands-on experience in tackling a real-world data science problem, offering insights into both the technical and operational challenges of working with complex, real-life datasets. Throughout the project, we gained a deeper understanding of how critical communication and collaboration are between technical teams and domain experts. These insights were instrumental in ensuring the success of the project and improving our problem-solving approach.

The primary objective of the project was to develop a ML-based solution to benchmark the maintenance efficiency of hydropower plants. Statkraft had long recognized the need for a tool that could evaluate plant efficiency across different asset classes, regions and business-level groupings, while helping them to be more efficient in maintenance management. The overarching goal was to provide the company with insights into the performance of their assets, identifying underperforming plants, areas of inefficiency, and the best practices that could be shared across the organization.

Our project set out with clear goals: (1) to provide a maintenance-based analysis of historical data from 2021–2024, (2) to develop a ML model that quantifies plant efficiency based on maintenance performance, (3) to deliver a tool for comparing MBE across hydropower plants, and (4) to enable benchmarking across plants, business groups, and regions.

We believe that we successfully met these objectives. Our analysis of historical maintenance data resulted in a detailed understanding of maintenance performance and operational inefficiencies. We developed a ML model to quantify MBE, and the model demonstrated strong predictive capabilities, allowing for both plant-specific time series predictions and aggregate efficiency scores. The selected model, Random Forest Regressor, provides a solid foundation for further refinement and scaling, especially when actual operational data becomes available.

Moreover, the Power BI visualizations for benchmarking tool we delivered meets Statkraft's need for comparing plant performance across different asset classes and regions. The interactive visualizations allow users to identify plants with suboptimal performance, track maintenance cost efficiency, and make data-driven decisions about resource allocation. This aligns closely with Statkraft's objective to gain better control of their maintenance strategies and to benchmark their assets more effectively.

However, while we believe that we have successfully addressed the core goals, there are still areas for improvement. A key limitation of the project is that the models were trained using synthetic data, in which case while useful for prototyping and testing, it may not fully capture the complexity and nuances of real-world operational data. The accuracy of the ML model is directly tied to the quality and relevance of the training data. Therefore, further work is needed to enhance the synthetic data's complexity, ensuring it more closely aligns with actual operational conditions. Additionally, the integration of the model with SCADA systems for real-time data collection and analysis would be a significant next step. Real-time predictions and early detection capabilities could vastly improve the practicality of the system, enabling proactive maintenance and more immediate operational adjustments.

This project has also provided us with crucial lessons on the importance of cross-functional collaboration. Working closely with Statkraft's domain experts allowed us to better understand the operational context and ensure that our solution would meet their real-world needs. Going forward, integrating user feedback, improving data quality, and expanding model capabilities are key steps to ensuring the long-term success of this solution.

In conclusion, while there are areas that require further refinement and validation, this project has successfully delivered a tool that meets Statkraft's immediate goals and lays the groundwork for future enhancements. We have provided a solid, scalable solution that not only benchmarks plant performance but also enables proactive maintenance planning, aligning closely with Statkraft's long-term vision.

7 Glossary

In order to provide clarity and consistency throughout this report, a glossary of key terms, concepts and abbreviation related to the subject matter are listed below.

- **Abrasion:** Damage results from hard particles or sediments physically wearing down a surface through friction.
- **Asset Class:** Classification system used by Statkraft to indicate the operational importance of hydropower plants.
- **Benchmarking:** A comparison method to evaluate the performance and maintenance efficiency of different hydropower plants.
- **Blades / Guide Vanes:** Turbine components that direct water flow toward the runner to optimize energy conversion.
- **Capacity Bins:** Categorization of power plants into groups by installed capacity.
- **Cavitation:** Refers to the physical damage caused by the formation and collapse of vapor bubbles in a fluid, typically occurring on turbine blades, leading to pitting and surface erosion.
- **Constructed Efficiency (CE):** Target feature, the values being trained for within the synthetic dataset.
- **Cost Generating Units (CGU):** Refers to Cost Generating Units, a grouping within Statkraft that represents a collection of interconnected hydropower plants
- **Data Imputation:** The process of replacing missing or incomplete data with substituted values to create a complete dataset suitable for analysis or modeling.
- **Data Leakage:** A situation where information outside the training dataset is inadvertently used during the training process, giving the model access to future or unintended data effectively allowing the model to "cheat".
- **Exploratory Data Analysis (EDA):** Statistical analysis and visualization performed to understand patterns in maintenance, downtime, and production data.
- **Feature Engineering:** The process of creating meaningful input variables from raw data to improve model performance to better represent the underlying patterns and relationships in the data.
- **Generalization:** A model's ability to make accurate predictions on new, unseen data.
- **Hyperparameter Tuning:** Selecting the best model settings (e.g., tree depth, learning rate) to improve prediction accuracy.
- **Installed Capacity:** The plants' total electrical generation capacity measured in megawatts (MW) or gigawatts (GW).
- **Lasso (Least Absolute Shrinkage and Selection Operator):** A regression method that performs variable selection and regularization by shrinking some coefficients to zero, helping to simplify models and prevent overfitting.

- **Loss of Income (LINC):** Estimated financial loss from power plant unavailability, based on market electricity prices and unrealized energy production.

- **Machine learning (ML):** A field of artificial intelligence that enables systems to learn patterns from data and make predictions or decisions without being explicitly programmed.

- **Maintenance-Based Efficiency (MBE):** A metric that evaluates power plant performance based on the effectiveness and impact of maintenance practices.

- **MW / MWh:** Megawatt (MW) is a unit of power. Megawatt-hour (MWh) measures energy production or consumption over time.

- **Outliers:** Data points that lie significantly outside the typical range of a dataset, often defined as values beyond 1.5 times the interquartile range or more than 2–3 standard deviations from the mean.

- **Overfitting:** A modeling issue where a model learns the details and noise in the training data to the extent that it negatively impacts its performance on new, unseen data, leading to poor generalization.

- **Predicted Efficiency (PE):** The output values from our models.

- **Power Plant Group (PPG):** A hierarchical and financial grouping of power plants, organized for managerial and operational purposes.

- **Ridge:** A regularization technique used in linear regression that adds a penalty proportional to the square of the magnitude of the coefficients to the loss function. Ridge shrinks coefficients toward zero to reduce model complexity and multicollinearity, improving the model's generalization to new data.

- **Runner:** The rotating component of a hydropower turbine that converts hydraulic energy into mechanical energy.

- **SAC (SAP Analytics Cloud):** A cloud-based data management and analytics platform by SAP that integrates business intelligence, planning, and predictive analytics.

- **SAP (Systems, Applications, and Products):** A global software company known for enterprise resource planning (ERP) solutions, including analytics, business operations, and data management tools.

- **SCADA:** A Supervisory Control and Data Acquisition system used for monitoring and controlling industrial processes and infrastructure.

- **Seed Dataset:** An initial set of data used to generate new synthetic data. It provides the base structure, characteristics, or distribution from which additional synthetic data is created.

- **Supervised Learning:** A type of machine learning where models are trained using input data with known target values.

- **Synthetic Data:** Statistically generated data used to simulate real-world scenarios for model training.

- **Target Variable:** The value a machine learning model is trained to predict.

- **Unsupervised Learning:** A type of machine learning that finds patterns in data without using known target values.
- **Weighted Quantile Sketch:** An algorithm used (notably in XGBoost) to efficiently compute approximate quantiles from weighted data, which is helpful for making optimal split decisions in decision trees when some data points carry more importance than others.

8 Appendix

8.1 XGBoost Box plot: Predicted Efficiency Distribution

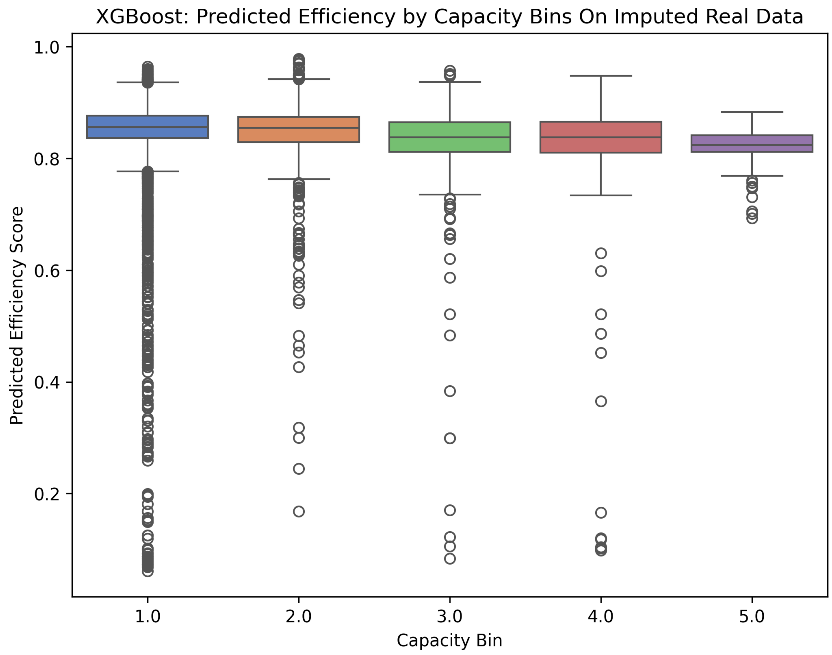


Figure 23: XGBoost: Predicted Efficiency Distribution

8.2 Random Forest Box plot: Predicted Efficiency Distribution

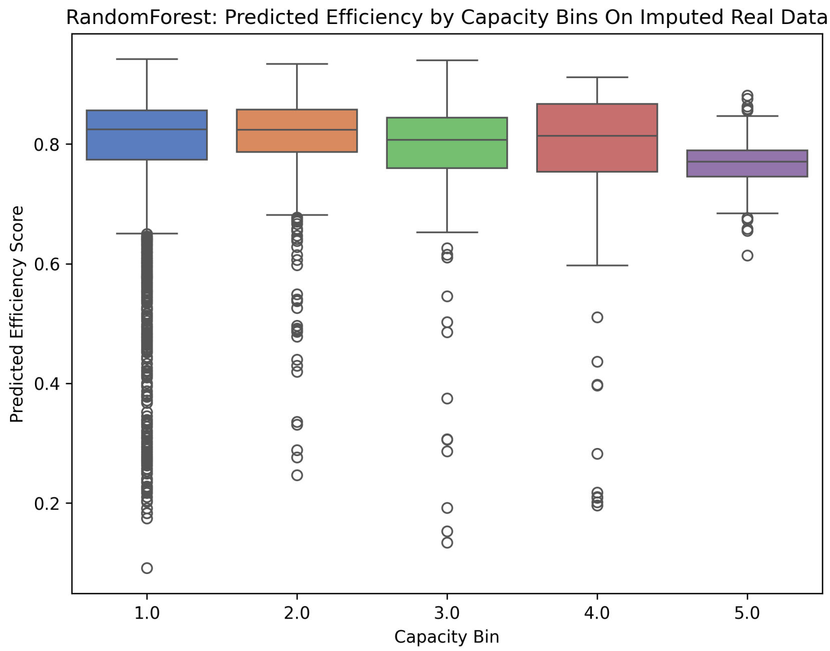


Figure 24: Random Forest: Predicted Efficiency Distribution

8.3 AutoEncoder Box plot: Predicted Efficiency Distribution

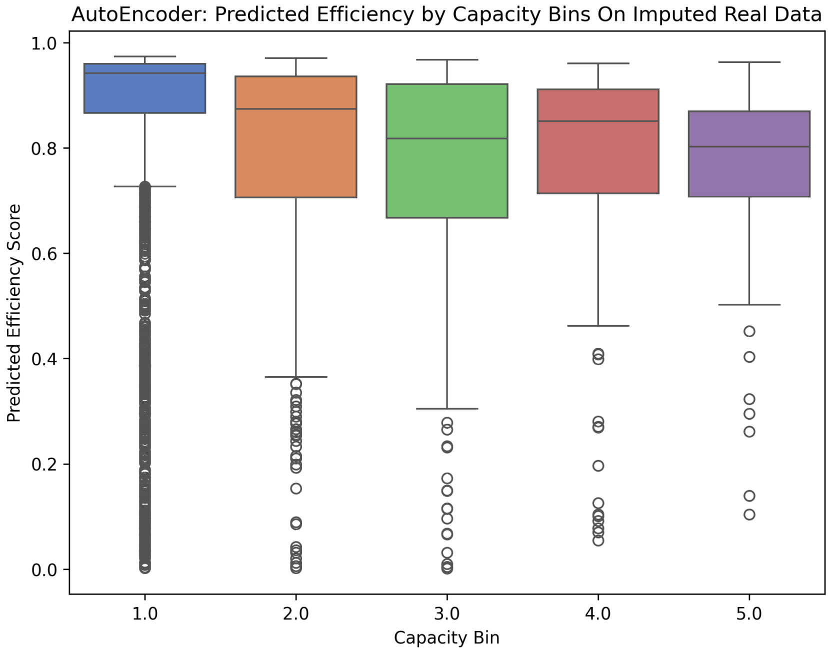


Figure 25: AutoEncoder: Predicted Efficiency Distribution

8.4 MLP Box plot: Predicted Efficiency Distribution

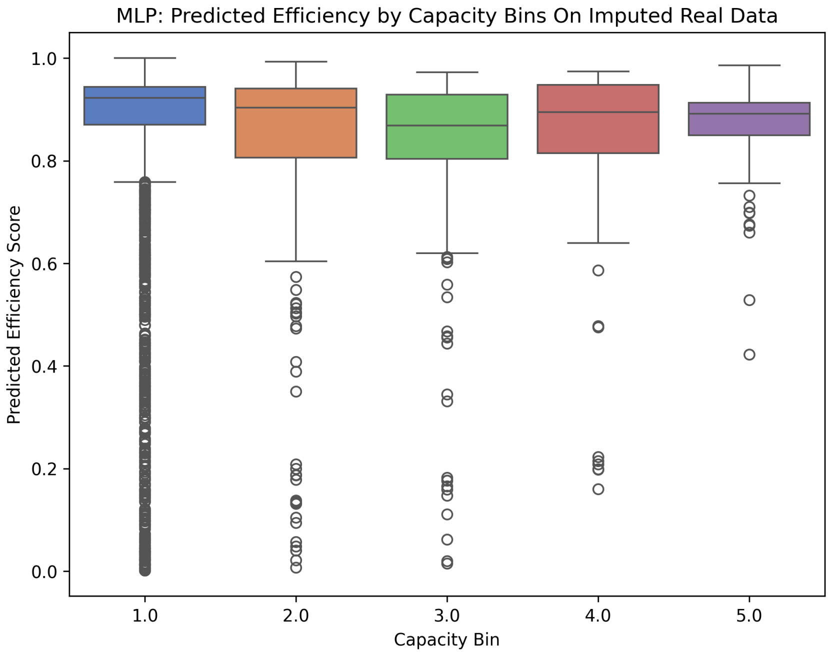


Figure 26: MLP: Predicted Efficiency Distribution

8.5 Seed vs Synthesized: distributions

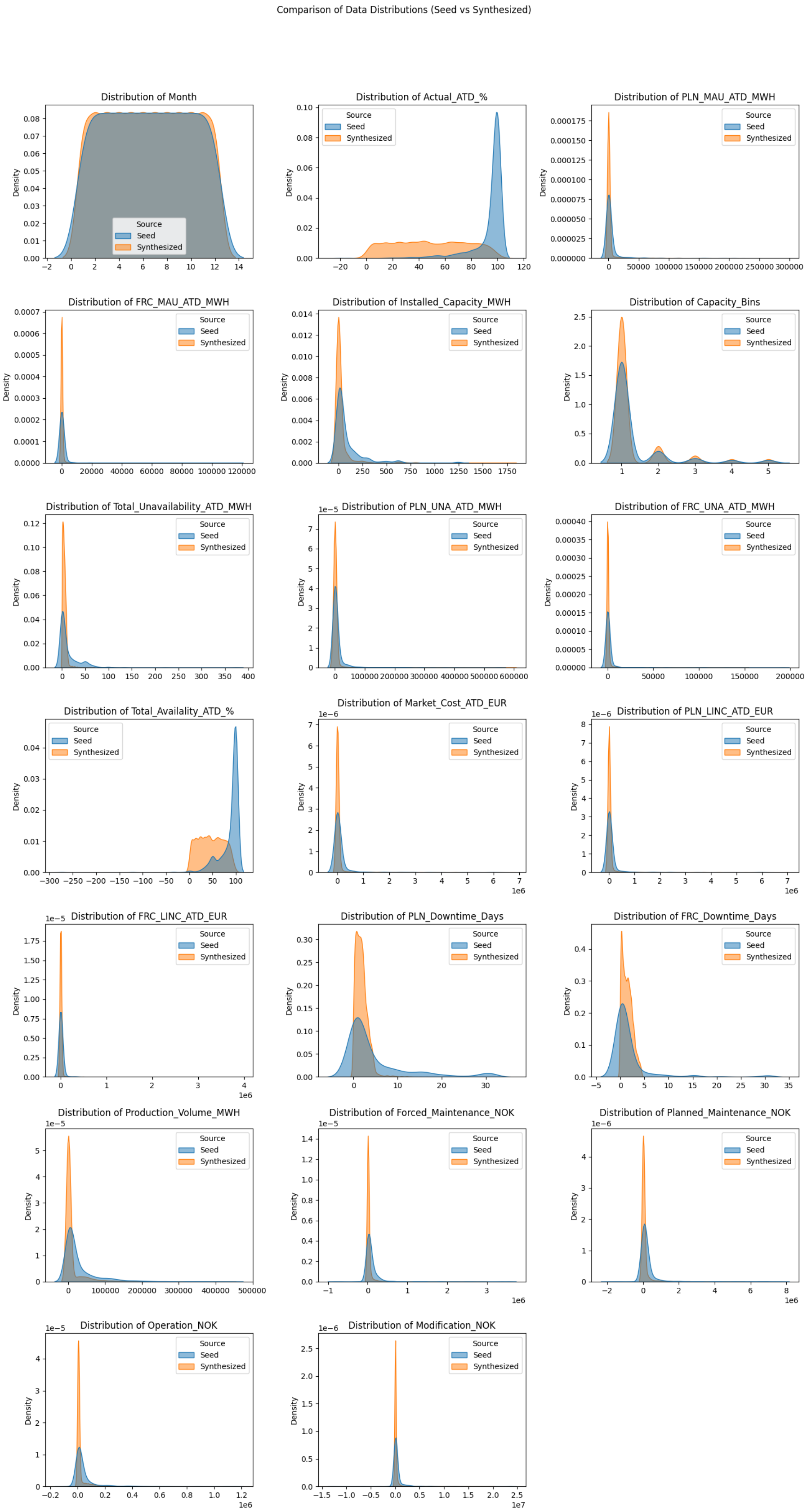


Figure 27: Seed VS Synthetic Data Distribution

8.6 Seed vs Synthesized: Data Pattern Comparisons

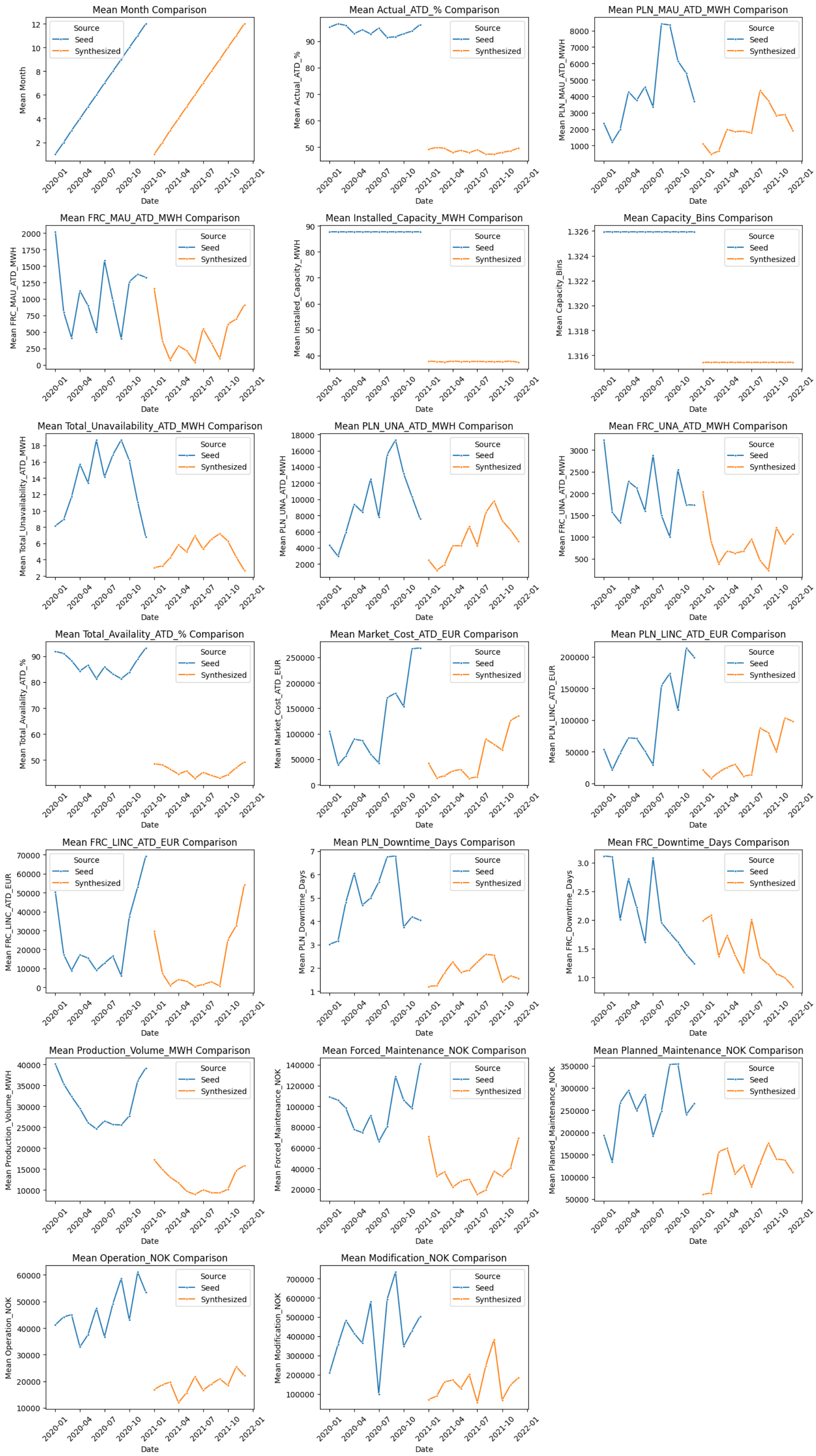


Figure 28: Seed vs Synthesized: Data Pattern Comparisons

8.7 Seed vs Synthesized: Violin distributions

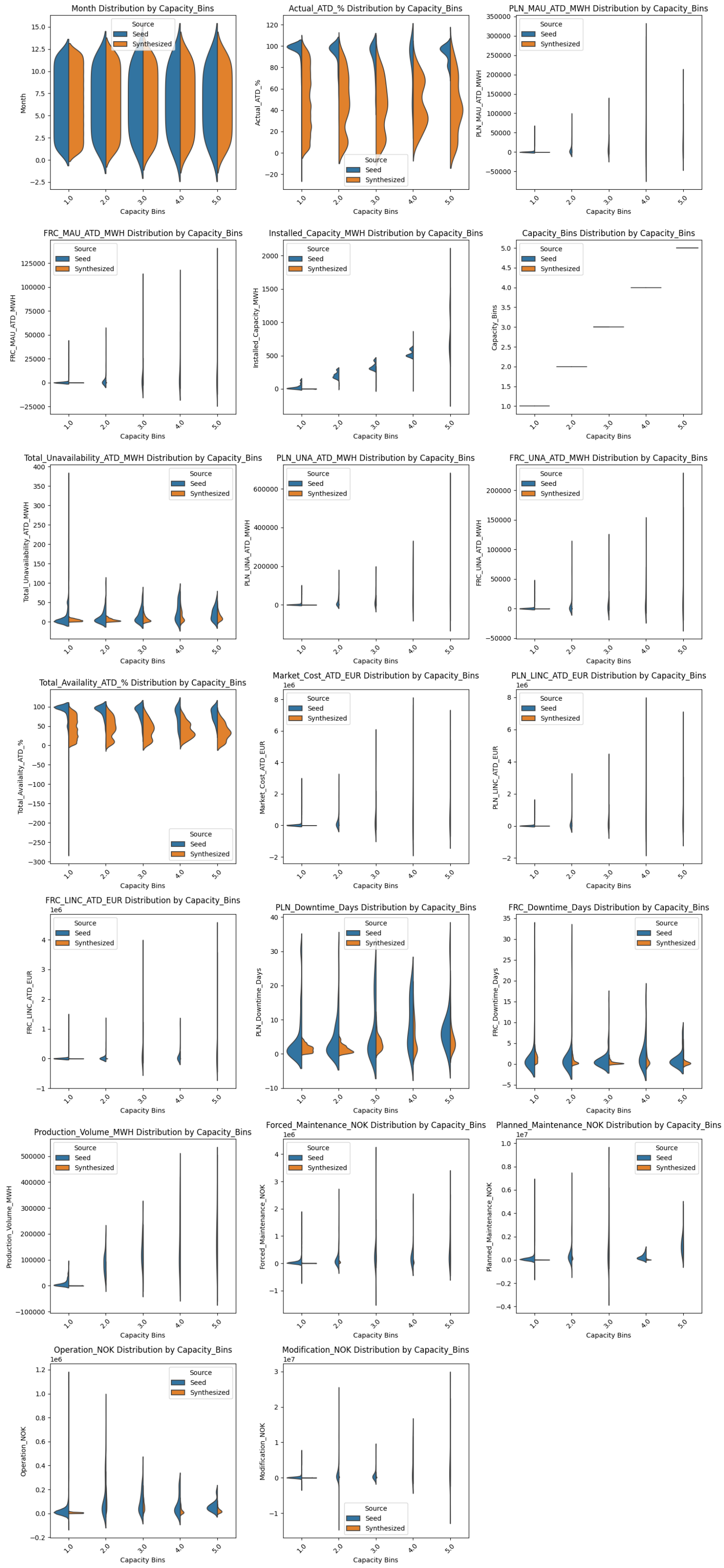


Figure 29: Violin: Distributions

References

[1] Akankunda, B., Adaramola, M. S., Nkundabanyanga, S. K., & Mukisa, N. (2024). *Techno-economic performance assessment of selected power plants in Uganda. Renewable Energy*, 101, 452-460. Retrieved from https://www.tandfonline.com/doi/full/10.1080/14786451.2024.2350980#abstract (Accessed: Apr 25, 2025)

[2] ASQ. (2024). *Benchmarking*. Retrieved from https://asq.org/quality-resources/benchmarking/ (Accessed: March 23, 2025)

[3] Bortoluzzi, M., Furlan, M., & Reis Neto, J. F. (2022). *Assessing the impact of hydropower projects in Brazil through data envelopment analysis and machine learning. Renewable Energy*, 179, 736-746. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S0960148122015609 (Accessed: Apr 25, 2025)

[4] Caballar, R. D., & Stryker, C. (2024). *8 best practices for synthetic data generation*. IBM. Retrieved from https://www.ibm.com/think/insights/synthetic-data-generation (Accessed: Apr 27, 2025).

[5] Camcode. (2025). *What is corrective maintenance?*. Retrieved from https://www.camcode.com/blog/what-is-corrective-maintenance/ (Accessed: March 28, 2025)

[6] Camp, R. C. (1989). *Benchmarking: The search for industry best practices that lead to superior performance*. Quality Press. Retrieved from https://archive.org/details/benchmarkingsear00camp

[7] National Center for Biotechnology Information. (2024). *Common Failures in Hydraulic Kaplan Turbine Blades and Practical Solutions*. Retrieved from https://pmc.ncbi.nlm.nih.gov/articles/PMC10179411/ (Accessed: March 23, 2025)

[8] U.S. Department of Energy. (2024). *How Hydropower Works*. Retrieved from https://www.energy.gov/eere/water/how-hydropower-works (Accessed: Feb 28, 2025)

[9] U.S. Department of Energy. (2024). *Types of hydropower turbines*. Retrieved from https://www.energy.gov/eere/water/types-hydropower-turbines (Accessed: March 23, 2025)

[10] Global Benchmarking Network. (2025). *Types of Benchmarking*. Retrieved from https://www.globalbenchmarking.org/index.php/whats-benchmarking/types-of-benchmarking/ (Accessed: May 8, 2025)

[11] Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press. Retrieved from http://alvarestech.com/temp/deep/Deep%20Learning%20by%20Ian%20Goodfellow,%20Yoshua%20Bengio,%20Aaron%20Courville%20(z-lib.org).pdf

[12] GridPro. (2024). *Understanding the flow through Francis turbines*. Retrieved from https://blog.gridpro.com/understanding-the-flow-through-francis-turbines (Accessed: March 23, 2025)

[13] Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann. Retrieved from https://myweb.sabanciuniv.edu/rdehkharghani/files/2016/02/The-Morgan-Kaufmann-Series-in-Data-Management-Systems-Jiawei-Han-Micheline-Kamber-Jian-I -Concepts-and-Techniques-3rd-Edition-Morgan-Kaufmann-2011.pdf (Accessed: March26, 2025)

[14] IBM. (2025). *Predictive vs preventive maintenance*. Retrieved from https://www.ibm.com/think/topics/predictive-vs-preventive-maintenance (Accessed: March 28, 2025)

[15] Jad, S., Desforges, X., Villard, P.-Y., Caussidéry, C., & Medjaher, K. (2025). *Diagnostic method for hydropower plant condition-based maintenance combining autoencoder with clustering algorithms*. arXiv. Retrieved from https://arxiv.org/pdf/2504.03649 (Accessed: Apr 27, 2025).

[16] Kim, S. (2024). *Optimal preventive maintenance: Balancing reliability and costs in the electricity market*. *Energy*, Volume 29, Issue 10, Pages 1615-1623. Retrieved from https://www.sciencedirect.com/science/article/abs/pii/S0301421524003367 (Accessed: March 25, 2025)

[17] Kumar, K., & Saini, R. P. (2022). *Economic analysis of operation and maintenance costs of hydropower plants*. Sustainable Energy Technologies and Assessments, 53, 102704. https://www.sciencedirect.com/science/article/pii/S2213138822007524(Accessed:March25,2025)

[18] Kundu, S., Saha, S., & Paul, S. (2022). *Hydropower: An efficient renewable source of energy – An analysis*. *International Journal of Engineering Research and Technology*, 11(3), 219–222. Retrieved from https://www.researchgate.net/publication/359577436_Hydropower_an_efficient_renewable_source_of_energy_An_analysis (Accessed: March 25, 2025)

[19] LibreTexts. (2024). *The Kaplan and Francis turbines*. Retrieved from https://eng.libretexts.org/Sandboxes/jhalpern/Energy_Alternatives/05%3A_Hydropower/5.03%3A_Methods_of_Harnessing_Hydropower/5.3.01%3A_The_Kaplan_and_Francis_Turbines (Accessed: March 23, 2025)

[20] Jardine, A. K. S., & Tsang, A. H. C. (2015). Maintenance management models: A study of the published literature to identify empirical evidence. A greater practical focus is needed. *ResearchGate*. Available at: https://www.researchgate.net/publication/279174646_Maintenance_management_models_a_study_of_the_published_literature_to_identify_empirical_evidence_A_greater_practical_focus_is_need. Accessed: March 26, 2025.

[21] Liaw, A., & Wiener, M. (2002). *randomForest: Breiman and Cutler's Random Forests for Classification and Regression*. R package version 4.7-1. Retrieved from https://cran.r-project.org/web/packages/randomForest/ (Accessed: Apr 25, 2025).

[22] Retief, H., Garcia Andarcia, M., Dickens, C., & Ghosh, S. (2025). *Dam volume prediction model development using ML algorithms*. arXiv. Retrieved from https://arxiv.org/abs/2502.19989 (Accessed: Apr 30, 2025).

[23] Ridha, W. K. M., Kashyzadeh, K. R., & Ghorbani, S. (2023). *Common Failures in Hydraulic Kaplan Turbine Blades and Practical Solutions*. Materials, 16(9), 3303. Retrieved from https://doi.org/10.3390/ma16093303 (Accessed: March 26, 2025).

[24] Simscale. (2024). *Pelton turbine*. Retrieved from https://www.simscale.com/blog/pelton-turbine/ (Accessed: March 23, 2025).

[25] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2024). *sklearn.ensemble.RandomForestRegressor — scikit-learn 1.4 docu-

mentation. Retrieved from https://scikit-learn.org/stable/modules/ensemble.html (Accessed: Apr 25, 2025).

[26] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2024). *sklearn.ensemble.RandomForestRegressor — scikit-learn stable documentation*. Retrieved from https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html (Accessed: Apr 25, 2025).

[27] Statkraft. (2024). *Annual Results 2024*. Retrieved from https://www.statkraft.ie/news/2025/annual-results-2024/ (Accessed: March 19, 2025)

[28] Statkraft. (2024). *Statkraft in Europe: A leader in renewable energy*. Retrieved from https://www.statkraft.com/markets/europe (Accessed: March 19, 2025)

[29] Statkraft. (2024). *Statkraft's investments in South America*. Retrieved from https://www.statkraft.com/markets/south-america (Accessed: March 19, 2025)

[30] Statkraft. (2024). *Statkraft inaugurates the company's largest wind farm outside Europe in Brazil*. Retrieved from https://www.statkraft.com/newsroom/news-and-stories/2024/statkraft-inaugurates-the-companys-largest-wind-farm-outside-europe-in-brazil (Accessed: March 19, 2025)

[31] Statkraft. (2024). *Statkraft's operations in Asia*. Retrieved from https://www.statkraft.com/markets/asia/ (Accessed: March 19, 2025)

[32] Statkraft. (2024). *Hydropower capacity in Norway and Sweden*. Retrieved from https://www.statkraft.com/about-statkraft/where-we-operate/norway/ (Accessed: March 23, 2025)

[33] Statkraft. (n.d.). *Hydropower*. Retrieved from https://www.statkraft.com/what-we-do/hydropower/ (Accessed: March 23, 2025).

[34] Technomax. (2025). *Condition-based vs predictive maintenance*. Retrieved from https://www.technomaxme.com/condition-based-vs-predictive-maintenance/ (Accessed: March 28, 2025)

[35] Chollet, F., & the TensorFlow team. (2023). *Autoencoder tutorial — TensorFlow*. Retrieved from https://www.tensorflow.org/tutorials/generative/autoencoder (Accessed: Apr 25, 2025).

[36] Abdelrahman, A. R. A., & Saber, H. M. A. (2021). *Studying the factors that affect the efficiency of water turbines*. Dakahlia STEM High School, Egypt. Scientific Research Journal, 9(5). DOI: 10.31364/SCIRJ/v9.i05.2021.P0521856

[37] Upkeep. (2025). *What is unplanned maintenance?*. Retrieved from https://upkeep.com/blog/what-is-unplanned-maintenance/ (Accessed: March 28, 2025)

[38] Wong, L., Liu, D., Berti-Equille, L., Alnegheimish, S., & Veeramachaneni, K. (2022). *AER: Auto-Encoder with Regression for Time Series Anomaly Detection*. arXiv. Retrieved from https://arxiv.org/pdf/2212.13558 (Accessed: Apr 25, 2025).

[39] World Bank. (2020). *Operation and Maintenance Strategies for Hydropower: Handbook for Practitioners and Decision Makers*. Washington, DC: World Bank. Retrieved from https://www.worldbank.org/en/topic/energy/publication/

operation-and-maintenance-strategies-for-hydropower (Accessed: Apr 30, 2025).

[40] XGBoost Developers. (2020). *XGBoost Documentation (Release 3.0.0)*. Retrieved from https://xgboost.readthedocs.io/en/release_3.0.0/ (Accessed: Apr 25, 2025).

[41] Zolfaghari, M., & Golabi, M. R. (2021). *Modeling and predicting the electricity production in hydropower using conjunction of wavelet transform, long short-term memory, and random forest models*. Renewable Energy, 170, 1367–1381. Retrieved from https://doi.org/10.1016/j.renene.2021.02.017 (Accessed: Apr 30, 2025).